

HPS-DETR: Enhancing Small Object Detection With Lightweight Feature Extraction and Transformer Integration

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Abstract—Remote sensing visible-light small object detection is a challenging and common problem. Due to persistent objective factors such as complex backgrounds, it is still difficult to balance the detection accuracy and the number of parameters for densely packed and partially occluded small objects. Existing small object detection methods have neglected the issue of synchronous growth in the number of parameters while improving accuracy. To address these issues, we propose the higher precision small object detection transformer (HPS-DETR), a model optimized for remote sensing imagery with improved precision and real-time capability with fewer parameters. HPS-DETR introduces four key innovations. First, the faster convolutional gated selection unit (Faster-CGSU) enhances multiscale feature extraction by integrating a gating mechanism with partial convolutions (PConvs), improving the aggregation of local and global features while reducing redundant parameters. Second, the attention-based intrascale feature interaction using cascade group attention (CGA-AIFI) module focuses on critical feature regions while suppressing irrelevant background noise to improve attention diversity. Third, the cross-scale dynamic feature fusion (CDFF) module enriches small object detection through dynamic-scale fusion and precise feature encoding, mitigating feature loss during upsampling and downsampling. Finally, the Inner-MPDIoU loss function enhances bounding-box regression by prioritizing hard samples, improving accuracy, and convergence. The experimental results in SIMD, DOTA-v1.0, and NWPU VHR-10 datasets demonstrate mAP_{0.5} improvements of 2.4%, 1.7%, and 3.1%, respectively, with a 22.11% reduction in parameter count compared to the baseline. The AP_s also improves by 2.5%, 2.2%, and 3.3%, respectively. In WeatherDrone, VEDAI, and RTTS datasets, mAP_{0.5} increases by 1.9%, 3.8%, and 2.1%, demonstrating superior performance in small object detection and computational efficiency.

Index Terms—Feature fusion, high precision, real-time detection transformer (DETR), remote sensing images, small objects detection.

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I. INTRODUCTION

RAPID advancements in technology and the evolution of smart cities have underscored the importance of remote sensing object detection in various critical applications, including national defense, emergency response, urban planning, and resource exploration [1], [2]. However, due to the distances and angles of capture in remote sensing, densely distributed and partially occluded small objects are common in images, combined with the complex background, the detection task becomes even more challenging, leading to increased missed detection rates and false detection rates [3]. This is one of the key issues to be addressed in the field of remote sensing detection, improving the detection accuracy of small objects is crucial to advance the intelligence and practical applicability of remote sensing object detection, while also minimizing the model size as much as possible, facilitating its extension to real-world applications such as UAV surveillance and aerial reconnaissance. It is of great significance in accelerating the promotion of engineering applications.

Object detection [4], [5], [6] aims to identify, classify, and localize objects within images using bounding boxes and confidence scores. Early detection methods relied on handcrafted features, which lacked the representational capacity needed for robust performance [7]. The introduction of AlexNet in 2012 marked the beginning of a deep learning revolution in object detection, enabling algorithms to leverage hierarchical feature representations. Detection approaches can be broadly categorized into two-stage and one-stage paradigms. Two-stage methods, such as R-CNN [8] and Faster R-CNN [9], employ a region proposal step followed by feature extraction for classification and regression. Although these methods achieve high accuracy, their multistep architecture incurs significant computational overhead, limiting real-time applicability. In contrast, one-stage methods such as SSD [10] and YOLO [11] directly predict object categories and coordinates from the input image, offering faster inference speeds and end-to-end performance. However, one-stage approaches often suffer from an imbalance between positive samples and background regions, which can impede the detection of small and densely distributed objects.

Existing techniques [12] face the issue of inadequate feature extraction when applied to the detection of small dense and partially occluded objects in remote sensing. Although

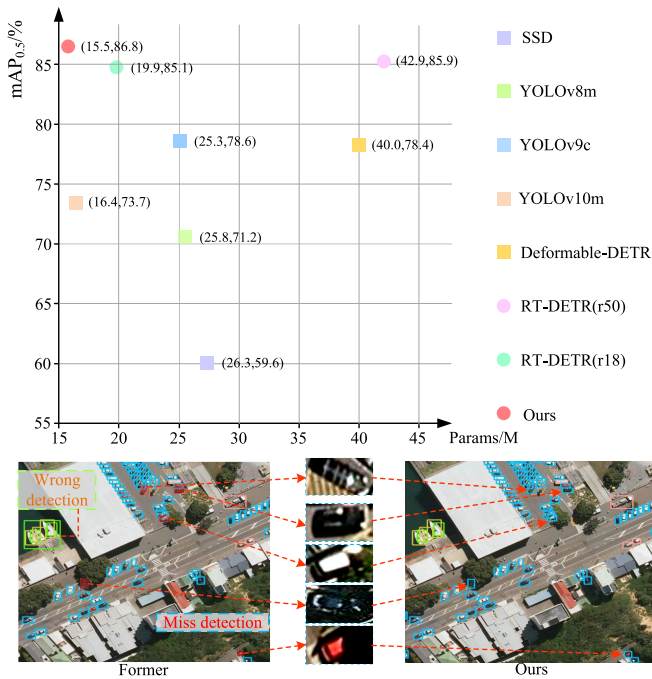


Fig. 1. Comparison plot of detection performance and parametric counts of different models tested on DOTA-v1.0 remote sensing image dataset. The top figure compares parametric count and accuracy between HPS-DETR (Ours) and some former classical object detection models. The bottom figure is a visual comparison of the detection results between one former model (here is the RT-DETR) and Ours, where objects misdetected or missed by the former model can be correctly detected by Ours.

convolutional neural networks (CNNs) are highly effective in extracting local features, their ability to capture global image context, such as long-range dependencies, remains limited. Although increasing the receptive field by stacking more convolutional layers alleviates this problem to some extent, it often complicates optimization and reduces model efficiency. Furthermore, overly deep models are not suitable for small object detection problems.

To address this limitation, transformers, originally designed for natural language processing tasks [13], have emerged as powerful tools to obtain long-range dependencies in vision tasks. Through self-attention mechanisms, the transformers excel at capturing global spatial relationships. However, their emphasis on global features often comes at the expense of local details preservation. So, these have motivated efforts to integrate CNNs and transformers, leading to the development of convolutional vision transformers that balance local feature extraction, global representations, and long-range dependencies. Detection transformer (DETR) [14], the first end-to-end transformer-based object detection framework, redefines object detection as a direct set prediction task. By eliminating nonmaximum suppression (NMS) and directly generating prediction boxes, DETR simplifies the detection pipeline. However, its high computational cost and slow convergence remain significant barriers to widespread adoption.

To overcome these challenges, several variants of DETR [15], [16], [17], [18] have been proposed and address many existing problems in the object detection task. However, these models still have issues such as poor adaptability

to small object scenarios, slow real-time performance, and large model parameters due to self-attention mechanisms. When it comes to small object detection in remote sensing images, dense distribution, partial occlusion, and complex backgrounds exacerbate these difficulties. Although methods such as FMSSD [19], YOLOv8 with small object detection structures [20], and transformer-based approaches [21], have achieved notable progress, issues such as feature compression during downsampling, ineffective multiscale fusion, and loss of small object details persist. Addressing these limitations is critical for advancing the state-of-the-art in remote sensing object detection.

RT-DETR [22] achieves real-time performance in object detection with reduced parameters and computational complexity by employing a hybrid encoder and intersection over union (IoU)-aware query selection. Building on the foundation of RT-DETR, our work aims to enhance the accuracy and efficiency of small object detection by striking a balance between local and global feature extraction while significantly reducing model parameters.

As illustrated in Fig. 1, the significant intent of HPS-DETR is to find a lightweight approach that enables a higher precision detection of small objects, which are densely or overlappingly distributed within complex backgrounds. The proposed faster convolutional gated selection unit (Faster-CGSU), a lightweight multiscale feature extraction module, extracts more effective feature information than conventional residual structures while reducing redundant information, increasing the amount of information/unit features. Its internal convolutional gated selection unit (CGSU) effectively enhances the passage of texture features of the objects, efficiently filtering out low-frequency features. By combining partial convolution (PConv) with a three-layer residual structure, the feature extraction capability is further strengthened. In addition, we introduce the cascaded group attention module (CGA-AIFI) to enhance the diversity of object features, thereby improving the ability of the model to detect objects of varying scales and spatial distributions. The cross-scale dynamic feature fusion (CDFD) structure can split, adjust, and fuse feature maps of different scales to enhance detailed feature representations. Internally, it employs a channel-parallel dynamic upsampling module to effectively learn feature information from different channels, efficiently fusing the high-dimensional information from deep features with the detailed information from shallow feature maps, and minimizing the loss of object information, especially small object features, during upsampling and downsampling processes. Finally, our proposed Inner-MPDIoU loss function incorporates auxiliary boundary supervision to compute losses and accelerate the regression of the bounding box, improving the ability of the model to focus on hard objects, such as overlapping or partially occluded ones, and enhancing the overall performance of the model in small object detection tasks.

Our main contributions are as follows.

- 1) We propose a lightweight multiscale feature extraction module, **Faster-CGSU**, leveraging the convolutional gated selection unit. This module enhances feature

extraction capability through a three-layer residual structure, reduces computational redundancy through a gating selection unit, and preserves fine-grained features crucial for small object detection, thereby improving real-time detection accuracy and reducing the number of parameters.

- 2) We present the **CGA-AIFI** module to enhance feature diversity and facilitate interactions on multiple scales, improving the detection of objects of varying sizes and spatial distributions.
- 3) We develop a **CDFF** module, which internally leverages a learned channel-parallel upsampling module to effectively utilize feature information from different channels. This module efficiently integrates features at various scales, mitigates information loss during up-sampling and down-sampling processes, and preserves crucial details of small objects, especially for densely overlapping ones.
- 4) We introduce the **Inner-MPDIoU** loss function, designed to better handle small objects that overlap and occlude. This enhancement not only boosts detection accuracy, but also accelerates model convergence.

To validate the effectiveness of HPS-DETR, extensive experiments are performed on SIMD [23], DOTA-v1.0 [24], and NWPU VHR-10 [25] three remote sensing image datasets. Our method achieves competitive detection results compared to current classic mainstream algorithms with reduced model parameters. Meanwhile, to verify the generalization and practical applicability of the model, we conduct validation experiments on the WeatherDrone [26], VEDAI [27], and RTTS [28] datasets. The results show that HPS-DETR improves detection accuracy by 1.9%, 3.8%, and 2.1% compared to the original model, respectively, fully demonstrating that HPS-DETR can be effectively applied to real-world scenarios such as UAV surveillance and autonomous driving.

The rest of this article is organized as follows. In Section II, we provide a brief review of CNN-based, transformer-based small object detection methods. In Section III, the overall structures of the proposed HPS-DETR and the Faster-CGSU, CGA-AIFI, CDFF, and Inner-MPDIoU modules are discussed in detail. In Section IV, we perform in-depth experiments and analyze the results. Finally, in Section V, the conclusion is given.

II. RELATED WORK

Small object detection constitutes a prevalent challenge within the domain of computer vision, which holds a significant place in this field. In the context of high-resolution remote sensing imagery featuring intricate backgrounds, small objects occupy a limited number of pixels, making them highly susceptible to background noise and pose significant challenges for feature extraction. Researchers have proposed a plethora of methodologies to tackle the challenges associated with small object detection in remote sensing, including advancements in feature extraction mechanisms, the incorporation of attention mechanisms, multiscale feature fusion techniques, and the introduction of novel loss functions. This

section provides a comprehensive review of mainstream object detection algorithms as well as recent advancements in small object detection.

A. CNN-Based Methods

Following the success of CNN-based methodologies in generic object detection, significant advances have been made to adapt these approaches to remote sensing imagery. Two-stage algorithms are known for their high detection accuracy. For example, cascaded R-CNN [29] refines predictions through a cascaded regression framework, enhancing detection precision and instance segmentation. AAF-Faster R-CNN [30] improves efficiency by incorporating additive activation functions (AAFs) using Fourier series, while SME-Net [31] improves feature maps with a split-merge strategy and corrects inconsistencies in spatial layout. LSKNet [32] dynamically adjusts receptive fields to address background disparities. Despite their accuracy, two-stage algorithms face challenges in speed and complexity. To address real-time detection demands, single-stage detectors have emerged. Attention and feature fusion SSD [33] enriches semantic content with multitier feature fusion and dual-path attention mechanisms. An enhanced YOLOv8s [34] reduces the parameters by integrating ghost blocks and replacing PAFPN with Bi-FPN. Furthermore, DCL-Net [35] improves feature discriminability and localization using a receptive field aggregation module and a bottom-up path aggregation module. However, CNN-based methods are inherently limited by the fixed receptive field of convolutional kernels, which impedes their ability to capture global features, undermining accuracy in object localization and classification.

B. Transformer-Based Methods

In recent years, transformer-based methods have made significant progress in object detection. Researchers have proposed various enhancement strategies to improve model detection accuracy and training efficiency. The deformable DETR [15] accelerates the convergence speed of the model and reduces algorithmic complexity by enhancing the efficiency of the attention mechanism while addressing the performance issues in small object detection by introducing multiscale features. Conditional DETR [16] improves convergence through conditional spatial queries, while anchor DETR [17] incorporates an anchor mechanism to streamline query optimization. To further enhance training efficiency, DAB-DETR [18] dynamically updates reference points to address training inefficiencies.

With the gradual application of the transformer model in remote sensing object detection, some researchers have started to explore how to further improve the detection accuracy of small objects by improving the model architecture. For example, an enhanced Swin transformer [36], incorporating a locally aware backbone, improves local feature perception and improves small object detection accuracy. TRD [37] represents an end-to-end transformer-based paradigm, which uses an attention-laden backbone to increase detection performance. SFRNet [38] utilizes discrete feature refinement, spatial and channel transformers, and a multi-ROI loss function to extract

fine-grained classification differences and rotation-sensitive features. Reference [39] improves RT-DETR by adopting grouping and cascading strategies to improve the use of features, achieving robust detection in complex environments.

Although these transformer-based approaches have significantly improved detection performance, they face challenges in detecting tiny objects, often burdened by redundant parameters.

C. Small Objects Methods

In small object detection, CGMDet [40] integrates positional information with enhanced regression loss of the bounding box using a coordinate and global information aggregation module (CGAM) and a feature fusion module (FFM). Drone-YOLO [41], with a three-layer PAFPN architecture and a specialized detection head, improves the small object detection capabilities. ESOD [42] combines feature-level object search with patch splitting and sparse detection to minimize computational redundancy. SADENet [43] employs a sample vertebral structure and an enhanced contextual attention module (ECAM) to improve accuracy in complex environments. A feature pyramid composite neural network [44] introduces a context enhancement module (CEM) and a feature purification module (FPM) to mitigate conflicting information and boost performance. FFCA-YOLO [45] improves sensitivity to local regions, multiscale feature fusion, channel and space dimension using techniques such as a feature enhancement module (FEM) and a spatial context awareness module (SCAM).

GMDRA-net [46] proposes the global multilevel perception (GMP) module and the dynamic region aggregation (DRA) module, which enhance global perception capabilities and selectively aggregate effective features while filtering out noise, thus improving the recognition ability for small objects. GALDET [47] introduces a small object detection method based on global and local attention (GAL) mechanisms, which combines a novel FFM with a decoupled prediction structure to improve the detection accuracy of objects at different scales. It also optimizes the bounding box regression using the ZIoU loss function to enhance the localization accuracy for small objects. SODNet [48] proposes a small object detection algorithm based on adaptive spatial parallel convolution (ASP-Conv) and fast multiscale fusion (FMF), enhancing spatial feature extraction and cross-scale feature fusion to improve the accuracy and real-time performance of small object detection. These efforts optimize the overall model architecture from feature extraction, multiscale feature fusion, attention mechanism, and loss function with the purpose of improving the accuracy of small object detection.

Although these studies have markedly advanced small object detection in remote sensing images, challenges such as missed detections and false positives persist in scenarios characterized by complex backgrounds and dense arrangements. Inspired by prior work, we propose HPS-DETR, a higher precision small object DETR aimed at improving detection accuracy and reducing model parameters.

III. METHODS

In this section, we provide a thorough and comprehensive overview of the proposed HPS-DETR. First, we delineate the overall architecture and subsequently delve into a detailed elaboration of the composition and the underlying mechanisms of each individual module.

A. Overall Framework of HPS-DETR

The general architecture of HPS-DETR is built on RT-DETR (r18), the recently popular structure [15], [16], [17], [18], [22], and includes a backbone, a hybrid encoder, and a decoder with auxiliary detection heads. The hybrid encoder that can significantly improve inference speed by combining the intrascale and crossscale feature fusion consists of two components. One is attention-based intrascale feature interaction using cascade group attention (CGA-AIFI), and the other is CDFE.

The architecture of the HPS-DETR is shown in Fig. 2. To begin with, after three CBS processing in the backbone, the remote sensing image feature maps are carried out by the lightweight multiscale feature extraction module, Faster-CGSU, extracted features and generated multiscale feature maps P2, P3, P4, P5 in the hybrid encoder. Through the CGA-AIFI module, the hybrid encoder transforms the multiscale features of stage P5 into a sequence of image features and then gets the intrascale feature interaction information, named stage F5.

In the CDFE section, feature fusion is performed through three main paths. The first path involves the fusion of feature maps P3, P4, P5 using the designed dynamic scale sequence FFM (DSSFF). DSSFF mainly refines the extraction capability of channel features through a dynamic channel learning mechanism, which will be detailed subsequently.

The second path involves the feature maps P2, P3, P4 and the feature maps P3, P4, P5, being processed by two modules of triple feature encoder (TFE) [49], which are concatenated and fused along the channel dimension. The primary function of TFE is to capture detailed information about small objects by combining feature maps of different sizes, thereby improving the detection performance for densely packed small objects.

The third path performs the fusion of the F5 feature map with the output of the TFE modules, with a PAFPN [50] structure, achieving comprehensive integration of localization and semantic information through top-down and bottom-up cross-scale interactions, while the Fusion [22] block is to fuse two adjacent scale features into a new feature, having the input and output feature maps of the same dimensions.

In the decoder section, an IoU-aware query selection module is first used to select a certain number of image features as the initial object queries from the output sequence of the hybrid encoder. These queries are then optimized to generate predicted bounding boxes and confidence scores.

B. Faster Convolutional Gated Selection Unit

In remote sensing images, the presence of numerous small and densely occluded objects significantly increases the

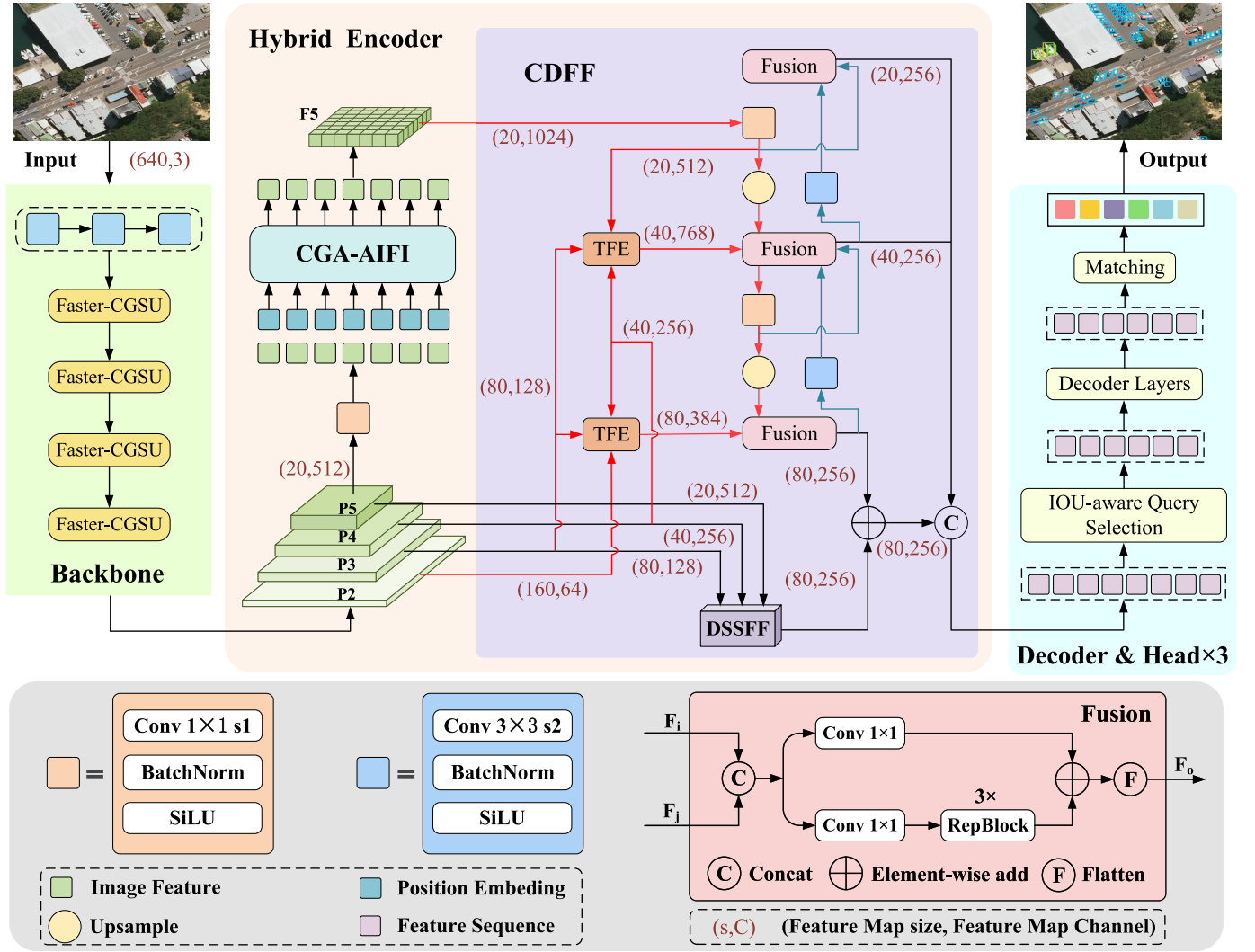


Fig. 2. Overall architecture of HPS-DETR. It includes a backbone network, a hybrid encoder, and a transformer decoder with three auxiliary detection heads. The four Faster-CGSUs within the backbone are the feature extraction modules. The hybrid encoder is used for feature fusion, consisting of the CGA-AIFI and the CDFF structure. The decoder with auxiliary prediction heads is used to generate predicted bounding boxes and confidence scores.

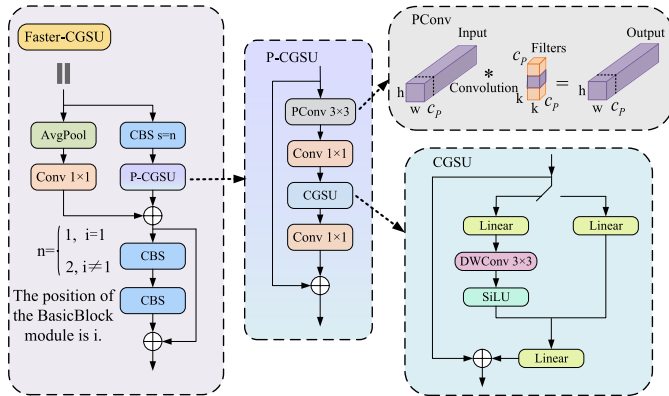


Fig. 3. Structure of the Faster-CGSU module. The main module of it is the partial convolutional gated selection unit (P-CGSU). The P-CGSU is a residual architecture with a forward path consisting of a PConv, two 1×1 convolutions, and a CGSU module. The PConv just processes the top and bottom information layers. The CGSU module has a selection mechanism that can reduce the number of redundancy parameters.

Although increasing the parameters typically leads to the improvement of detection accuracy, balancing the tradeoff between detection accuracy and model complexity is significant. So we design the Faster-CGSU lightweight multiscale feature extraction module, which employs a multilayer residual mechanism to achieve efficient gradient flow, enhancing non-linearity and abilities to extract spatial features. Our module integrates PConv from FasterNet [51] and the CGSU we design, improving the original feature extraction abilities and reducing redundancy parameters.

The structure of the Faster-CGSU module is shown in Fig. 3. The feature maps are produced through two paths and then be fused by element-wise add. The purpose of these is to extract the global and local information and then feed them into a residual network for further feature extraction.

The partial-CGSU (P-CGSU) is a crucial link in the Faster-CGSU and mainly consists of the PConv and the CGSU. The PConv operation performs computations by accessing memory, as lower-level feature maps typically contain more positional information, while higher-level feature maps focus more on the semantic information of the objects. To maintain

difficulty of accurate detection, and enhancing the feature extraction capability of the object detection model is necessary.

semantic consistency and enhance the ability of the model to represent object positions, we select the first and last consecutive channels c_p from the input feature map as representative feature maps for computation while the other channels remain unchanged. This ensures that the input and output feature maps maintain the same channels while significantly reducing computational complexity and memory access. The FLOPs and memory access of PConv are shown in the following equation:

$$h \times w \times k^2 \times c_p^2 \quad (1)$$

$$h \times w \times 2c_p + k^2 \times c_p^2 \approx h \times w \times 2c_p. \quad (2)$$

When selecting channel number c_p as 1/4 of the input channel number C , the computational complexity is reduced to 1/16 of the original standard convolution, and the memory access is reduced to 1/4 of the original standard convolution.

Unlike FasterNet [51], we design the CGSU to replace the BN layer after the 1×1 convolution to capture more local features of low-frequency texture, revealing its shortcomings in the detection of small objects within remote sensing images. The gating signal in the CGSU is controlled by the output of the activation function of the preceding layer. When the output parameters are sparse or almost zero, the right path is selected. This choice reduces one convolutional operation and introduces stochastic residuals through a linear transformation, thereby enhancing the generalizability of the model. In contrast, the model proceeds along the left path to further extract and aggregate information. The gating mechanism in CGSU reduces redundant data processing by minimizing the number of parameters and computational complexity, enabling the model to focus on useful feature information. Through depth-wise convolution (DWConv) [52], CGSU can more effectively capture local features, enhancing the ability of the model to process subtle details. The combination of convolution with the gating mechanism allows the model to retain the advantages of convolutional local feature extraction while effectively aggregating global information through linear transformation. In our experimental section, we conduct comparative experiments in five lightweight backbones, including FasterNet, for feature extraction, demonstrating the appropriateness of Faster-CGSU.

The Faster-CGSU module utilizes a multilayer residual mechanism compared to traditional convolutional feature extraction. It demonstrates superior performance in several aspects, including more effective use of multichannel information, reduced parameter count and computational overhead, and maintained gradient flow. These advantages enable the model to focus more on the local detailed features of the object during feature extraction, which is beneficial for identifying and detecting dense and partially occluded objects.

C. Attention-Based Intrascale Feature Interaction Using Cascade Group Attention Module

Due to the small size and dense occlusion of remote sensing objects, it is crucial to extract the visible features that remain in the regions of unoccluded objects. Feature extraction needs further refinement to prevent false negatives or positives. The introduction of the CGA-AIFI enhances the diversity

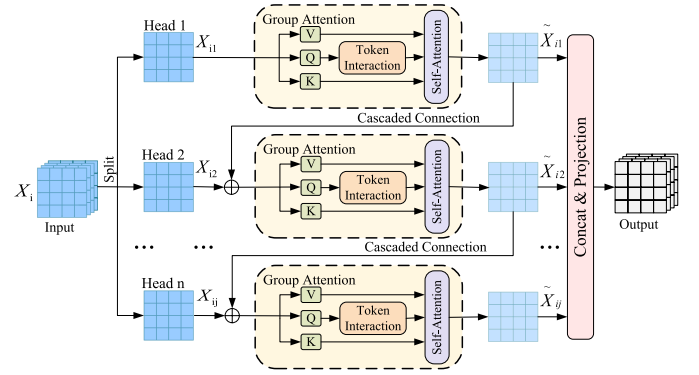


Fig. 4. Structure of the CGA-AIFI module. The feature map inputs are split into groups and each group matches each subspace to learn distinctive slice features. The output from each group is cascaded to the next group. Finally, the outputs from all groups are concatenated as the final output.

of the attention mechanism, reducing the loss of effective information.

Based on the transformer, CGA-AIFI performs a deep interaction on the P5-level feature map, delve into the high-level semantic information, compensating for the insufficient global attention of the object during the previous feature extraction period. We introduce cascaded group attention (CGA) [53] to AIFI, effectively decomposing the complexity of the attention computation and refining different sliced cascade features, which is beneficial for achieving a more adequate balance between local and global features. The structure of the CGA module is shown in Fig. 4. Unlike multihead self-attention (MHSA), we do segmentation before calculating V, Q, and K, dividing the attention heads into multiple subattention heads. Since different heads focus on distinct local regions, they collectively achieve global semantic coverage from a holistic perspective, achieving an initial balance between local and global features. The input sequence of each group first undergoes linear mapping to generate the value (V), query (Q), and key (K). After Q, a token interaction layer is introduced to capture the subtle connections between local features, as well as the positioning and semantics of these local features within the global context. Then, CGA-AIFI applies grouped attention and uses V, Q, and K to calculate the weights of attention within each subspace to produce the output of grouped attention. In addition, by cascading the outputs of multiple different groups, diversified feature mappings can be learned. The local feature information of objects extracted by the preceding attention heads will be integrated during the computations of subsequent heads. As the heads are stacked layer by layer, the feature representations are continuously refined, preserving the details of local features while gradually incorporating the contextual information of global features. The cascaded outputs are then further concatenated as the final output. The introduction of CGA not only boosts the accuracy of object detection, but also lowers the space complexity, enhancing the ability of the model to perceive and accurately describe small object features. The specific computation process is as follows:

$$\tilde{X}_{ij} = \text{Attn}\left(X_{ij}P_{ij}^V, X_{ij}P_{ij}^Q, X_{ij}P_{ij}^K\right) \quad (3)$$

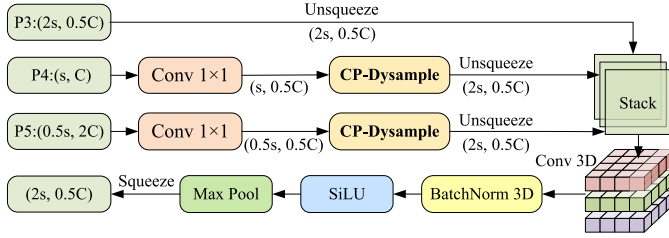


Fig. 5. Structure of the DSSFF module. s represents the feature map size, and C represents the number of channels. The channels and sizes of the P4 and P5 feature maps are adjusted by 1×1 convolution and CP-Dysample, and then the three-way feature maps are superimposed and activated by the 3-D convolution normalization operation so that they are fused into a $(2s, 0.5C)$ feature map.

$$X'_{ij} = X_{ij} + \tilde{X}_{i(j-1)}, \quad 1 < j \leq n \quad (4)$$

$$\tilde{X}_{i+1} = \text{Concat} \left[\tilde{X}_{ij} \right]_{j=1:n} W_i^p \quad (5)$$

where X_{ij} is the j th slice projection of the input feature X_i , $\tilde{X}_{ij}$ is the output of X_{ij} after passing through the j th attention head, P_{ij}^V , P_{ij}^Q , and P_{ij}^K represent the V, Q, and K projections of the input feature map at different layers, X'_{ij} represents the sum of the input of current layer and the output of previous layer, n is the total number of heads, and W_i^p is a linear layer that reprojects the concatenated output features back to the same dimension as the input feature.

D. Cross-Scale Dynamic FFM

CDFF performs the task of CDFF which is the core of the hybrid encoder. After several previous feature extraction operations, the set of feature maps P2, P3, P4, P5, and F5 contains various information and needs to be fused. By focusing on the differences and correlations of feature representations at different scales, CDFF effectively transfers and fuses the object location and semantic information, achieving the goal of fully utilizing the feature pyramid network for feature fusion.

Primarily, CDFF consists of the TFE and DSSFF modules. Due to the numerous small objects that densely overlap in remote sensing images, the TFE module [49] is used for image expansion to accurately detect small objects. The TFE module can adjust and merge feature maps of large, medium, and small sizes to enhance the detailed feature information of the objects. First, the TFE module adjusts the channel of large- and small-scale feature layers using 1×1 convolutions to ensure consistency with medium-scale features. Then, a hybrid structure of max pooling and average pooling is applied to downsample the large-scale feature map, increasing the diversity of semantic information. For small-scale feature maps, nearest-neighbor interpolation is used for upsampling, preserving the richness of local features in low-resolution images and preventing the loss of small object feature information. Finally, after nonlinear optimization, the large, medium, and small feature maps are concatenated along the channel dimension to fuse the features. This process is detailed in the following equation:

$$F_{\text{TFE}} = \text{Concat}(F_l, F_m, F_s) \quad (6)$$

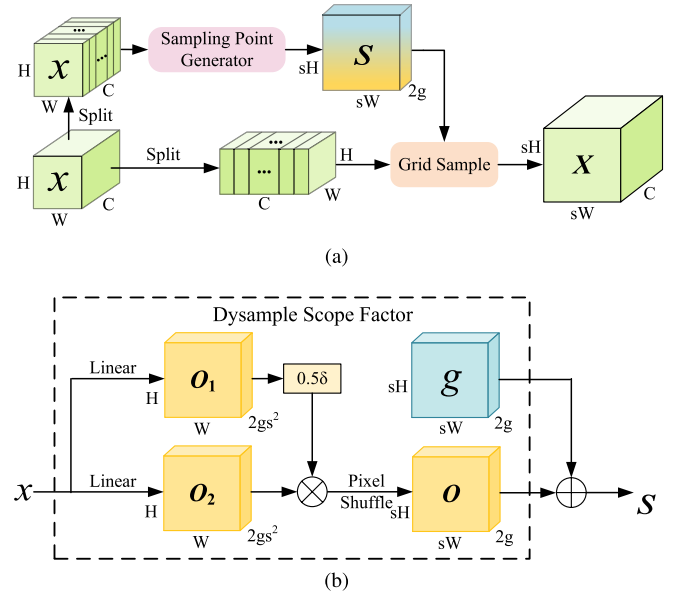


Fig. 6. Structure of the CP-Dysample module. (a) CP-Dysample process based on point sampling. The input feature map is split and then goes through the point sampling generator and point sampling set S to obtain the coordinates of the sampling points, and then the input feature map x is up-sampled in parallel by the grid sample function. (b) Dynamic point sampling generator. The point sampling set S is generated by combining the offset O with the original sampling grid g . δ is the dynamic range factor.

where F_{TFE} represents the feature map output from the TFE module, and F_l , F_m , and F_s represent large, medium, and small-scale feature maps. F_{TFE} is obtained by concatenating F_l , F_m , and F_s . F_{TFE} has the same resolution as F_m , and its number of channels is three times that of F_m .

We propose the DSSFF module, as shown in Fig. 5. DSSFF fully exploits the scale-space characteristics of the image. For the P4 and P5 level feature maps, the 1×1 convolution maintains scale invariance while adjusting their channel numbers. In addition, a channel parallel dynamic upsample (CP-Dysample) is employed to dynamically change the feature map size in parallel, extracting key information. The three feature maps with exact resolution are then concatenated and stacked, followed by 3-D convolutions to extract their scale-sequence features. DSSFF effectively integrates information from different feature maps, enhancing the feature fusion capability.

The CP-Dysample module adopts a learning-based dynamic fusion approach, which can adaptively adjust the fusion weights and methods between features of different scales based on the current input data characteristics. This allows for more precise and efficient feature fusion. Given that the high-resolution feature map P3 contains essential information required for small object detection, the CP-Dysample module modifies the processing mechanism of the original Dysample [54] module. Instead of operating on the feature map, it introduces a channel-parallel mechanism, which more effectively utilizes feature information from different channels, preventing feature omission.

The structure of the CP-Dysample parallel channel dynamic upsampling module is shown in Fig. 6(a) shows the overall process of parallel dynamic upsampling of channels based

on point sampling. The input feature map x of size $H \times W \times C$ is first split into channels, and each subchannel passes through the sampling point generator in parallel to determine the sampling region of the input feature map and then obtains the coordinates of the sampling points through the point sampling set S of size $sH \times sW \times 2g$, in which $2g$ is the coordinates of the x -axis and the y -axis. Then, the grid sample linear interpolation function is used to make the point sampling set resample the subchannels of the input feature map in parallel, generating a new feature map X of size $sH \times sW \times C$, and the computational process is shown in the following equation:

$$X = \text{Grid Sample}(x, S). \quad (7)$$

Fig. 6(b) describes in detail the dynamic point sampling set generation process. After splitting the channel, the feature map x produces two offsets, O_1 and O_2 , with dimensions $H \times W \times 2g \times s^2$ through two parallel linear layers. The sampling method for Offset O_1 is dynamic sampling with a sampling factor of 0.5δ , ensuring the flexibility of the sampling structure and restricting the appropriate sampling region. Here, $\delta \in [0, 1]$, it is adaptively adjusted during the training stage. The coefficient, 0.5, can prevent drastic jumps in sampling points, which could degrade the quality of upsampling. The experimental parts validate that the optimal performance setting is 0.5. With no scaling change to Offset O_2 to maintain the stability of the sampling structure. Then, combine the offsets O_1 and O_2 through pixel shuffle operations to reshape them into an offset O of $sH \times sW \times 2g$. Finally, the sampling set S is generated by combining the offset O with the original sampling grid g .

Unlike traditional static interpolation upsampling methods, CP-DySample dynamically generates the distribution of sampling points based on the input feature content. This dynamic content-driven adjustment of the sampling strategy is particularly beneficial in complex scenarios, as it improves the adaptability of the model to challenging terrain conditions. In addition, it helps prevent the potential loss of information about small object features during the upsampling process.

E. Inner-MPDIoU Loss Function

IoU is a widely used localization loss function in object detection tasks. The higher the IoU value, the more significant the overlap between the predicted and ground-truth boxes, resulting in a higher localization accuracy. This article proposes a novel Inner-MPDIoU loss function by combining the concepts of Inner-IoU [55] and MPDIoU [56]. Using auxiliary boundary computation to accelerate bounding-box regression, introducing a ratio factor to control the proportion of the auxiliary bounding boxes for detecting objects in remote sensing images. In addition, the similarity between the predicted and ground-truth boxes is compared using the distances between their top-left and bottom-right corners, further improving the accuracy of bounding-box regression. The definition of Inner-MPDIoU is shown in Fig. 7.

During the calculation process, the MPDIoU method directly minimizes the distances, d_1 and d_2 , between the top

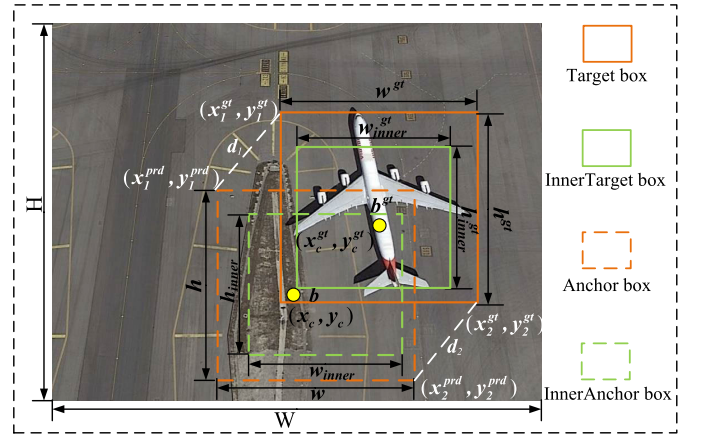


Fig. 7. Definition diagram of the Inner-MPDIoU loss function. The auxiliary box is precisely aligned with the center points of both the ground truth and predicted boxes. Using a scaling factor (ratio), the size of the auxiliary bounding box can be adjusted according to the actual scene requirements. In this article, the ratio is set to 0.95, less than 1, so the auxiliary bounding box size is smaller than that of the actual bounding box.

left and bottom right corners of the predicted and ground truth bounding boxes using the geometric properties of bounding box regression. This accelerates the similarity comparison between the two bounding boxes and effectively improves the convergence speed. During model training, each predicted bounding box $b = [x_c, y_c, w_{inner}, h_{inner}]$ is forced to move toward the ground truth box $b^{gt} = [x_c^{gt}, y_c^{gt}, w_{inner}^{gt}, h_{inner}^{gt}]$ to reduce the loss between the two bounding boxes. The Inner-IoU method uses a scaling factor, the ratio, to adjust the proportion of the auxiliary bounding box. More detailed image information is needed because the objects detected are objects in remote sensing images. After a series of experiments comparing different ratios, a ratio of 0.95 is chosen, which means that the auxiliary bounding box is smaller than the actual bounding box, thus reducing the effective prediction range compared to the IoU loss, as shown in Fig. 7. Although the gradient obtained from the auxiliary box has a more considerable absolute value than that from the IoU loss, it accelerates the convergence of high IoU samples and allows for more detailed image information by shrinking the size of the ground-truth bounding box. This setup effectively improves both the accuracy of the regression and the speed of convergence. Using the Inner-MPDIoU loss function further enhances the model by focusing on small objects without increasing computational or parameter requirements. The calculation process is as follows:

$$b_l^{gt} = x_c^{gt} - \frac{w^{gt} \times \text{ratio}}{2}, \quad b_r^{gt} = x_c^{gt} + \frac{w^{gt} \times \text{ratio}}{2} \quad (8)$$

$$b_t^{gt} = y_c^{gt} - \frac{h^{gt} \times \text{ratio}}{2}, \quad b_b^{gt} = y_c^{gt} + \frac{h^{gt} \times \text{ratio}}{2} \quad (9)$$

$$b_l = x_c - \frac{w \times \text{ratio}}{2}, \quad b_r = x_c + \frac{w \times \text{ratio}}{2} \quad (10)$$

$$b_t = y_c - \frac{h \times \text{ratio}}{2}, \quad b_b = y_c + \frac{h \times \text{ratio}}{2}. \quad (11)$$

Here, (8) and (9) calculate the left, right, top, and bottom boundaries of the truth bounding box. b_l^{gt} denotes the left boundary, which is obtained by subtracting the horizontal

coordinates of the center point of the real bounding box from its width w^{gt} multiplied by $1/2$ of the scaling factor ratio. Similarly, b_r^{gt} denotes the right boundary, obtained by adding the horizontal coordinates of the center point of the truth bounding box plus its width w^{gt} multiplied by the $1/2$ of the scaling factor ratio. b_l^{gt} and b_b^{gt} are obtained by similar calculations, with b_l^{gt} and b_b^{gt} denoting the top and bottom boundaries, respectively. Equations (10) and (11) calculate the left, right, top, and bottom boundaries of the anchor box using the same method

$$\text{inter} = (\min(b_r^{\text{gt}}, b_r) - \max(b_l^{\text{gt}}, b_l)) \times (\min(b_b^{\text{gt}}, b_b) - \max(b_t^{\text{gt}}, b_t)) \quad (12)$$

$$\text{union} = (w^{\text{gt}} \times h^{\text{gt}}) \times (\text{ratio})^2 + (w \times h) \times (\text{ratio})^2 - \text{inter} \quad (13)$$

$$\text{IoU}^{\text{inner}} = \frac{\text{inter}}{\text{union}}. \quad (14)$$

Equation (12) calculates the area of the overlap region between the ground truth box and the anchor box. Equation (13) calculates the area of the spliced area of the ground truth box and the anchor box. First, the area of the ground truth box and the anchor box are calculated, and both are multiplied by the square of the scaling factor ratio simultaneously. Then, the area of the stitching area is obtained by adding these two areas and subtracting the area of the overlapping area

$$d_1^2 = (x_1^{\text{prd}} - x_1^{\text{gt}})^2 + (y_1^{\text{prd}} - y_1^{\text{gt}})^2 \quad (15)$$

$$d_2^2 = (x_2^{\text{prd}} - x_2^{\text{gt}})^2 + (y_2^{\text{prd}} - y_2^{\text{gt}})^2 \quad (16)$$

$$L_{\text{Inner-MPDIoU}} = 1 - \left(\text{IoU}^{\text{inner}} - \frac{d_1^2 + d_2^2}{W^2 + H^2} \right). \quad (17)$$

Equation (17) calculates the loss of Inner-MPDIoU, where d_1 represents the distance between the top left corner of the predicted bounding box $(x_1^{\text{prd}}, y_1^{\text{prd}})$ and the top left corner of the truth bounding box $(x_1^{\text{gt}}, y_1^{\text{gt}})$, d_2 represents the distance between the bottom right corner of the predicted bounding box $(x_2^{\text{prd}}, y_2^{\text{prd}})$, and the top right corner of the truth bounding box $(x_2^{\text{gt}}, y_2^{\text{gt}})$. W and H represent the width and height of the input image, respectively.

IV. EXPERIMENTS AND RESULTS ANALYSIS

A. Datasets and Experimental Environment

1) *Datasets and Evaluation Metrics*: Considering the various categories of objects and the numerous small objects in remote sensing images, we perform ablation and comparative experiments on three public datasets, namely, SIMD [23], DOTA-v1.0 [24], and NWPU VHR-10 [25], to evaluate the performance of HPS-DETR.

The SIMD dataset, proposed by the National University of Sciences and Technology (NUST) of Pakistan, is a publicly available remote sensing image dataset. It contains 5000 remote sensing images with a resolution of 1024×768 pixels and 45 096 annotated instances, the majority of which are generic small vehicle objects with an average size of 23×23 pixels. The instances are categorized into

15 classes: car (C), truck (T), van (V), long vehicle (LV), bus (B), airliner (A), propeller airplane (PA), trainer aircraft (TA), charter aircraft (CA), fighter aircraft (FA), other (O), stair truck (ST), pulling truck (PT), helicopter (H), and ship (S). The dataset covers a wide range of complex scenarios, including urban roads, airports, and ports, with object distributions ranging from dense (with more than 100 objects/image) to sparse. It also includes various illumination and weather conditions. SIMD provides a high-quality dataset for evaluating the detection of dense small objects in remote sensing images that fits our study.

The DOTA-v1.0 dataset, proposed by Wuhan University and released in 2017, is widely used in remote sensing object detection and is known for its high resolution with a large number of densely distributed small objects, which poses significant challenges to detection algorithms. It consists of 2806 large-scale aerial images, with pixel sizes ranging from 800×800 to 4000×4000 , and contains 188 282 instances of varying scales and shapes. Among these, small object instances (smaller than 32×32 pixels) account for almost 60%. The instances are categorized into 15 classes: plane (PL), ship (SH), storage tank (ST), baseball diamond (BD), tennis court (TC), basketball court (BC), track field (TF), harbor (HA), bridge (BR), large vehicle (LV), small vehicle (SV), helicopter (HC), soccer ball field (SC), roundabout (RA), and swimming pool (SP). To preserve small object characteristics during feature extraction and down-sampling, images are segmented into 21 046 patches of 1024×1024 resolution using a sliding window with a stride of 824. Complex scenes in the dataset, such as crowded ports and airport aircraft clusters, thoroughly test the robustness of the model to detect small objects among scale variations and dense occlusions.

The NWPU VHR-10 dataset, introduced by Northwestern Polytechnical University in 2016, is a publicly available remote sensing dataset designed for spatial object detection, comprising 715 color images collected from Google Earth and 85 fused color infrared images obtained from the Vaihingen dataset with a resolution of 1024×1024 pixels. The spatial resolution of the images ranges from 0.08 to 2 m. The dataset includes a set of positive instances containing objects (650 images) and a set of negative instances without objects (150 images). The positive set contains 3651 annotated instances and is categorized into 10 classes: plane (PL), ship (S), storage tank (ST), baseball diamond (BD), tennis court (TC), basketball court (BC), ground track field (GTF), harbor (HA), bridge (BR), and vehicle (V). Balanced class distribution and diverse scene variations make it an ideal benchmark for evaluating the performance of HPS-DETR.

To comprehensively evaluate the effectiveness of the HPS-DETR algorithm for object detection in remote sensing images, we perform a comparative analysis with different algorithms. The evaluation is carried out using quantitative metrics, assessing the accuracy of the models by measuring the consistency between the detection results and the dataset labels. The experimental evaluation metrics include precision (P), recall (R), mean average precision (mAP), average precision for small objects (AP_s), average precision for medium objects (AP_m), average precision for large objects (AP_l),

the number of parameters (Param), and frames per second (FPS) for detection. Specifically, two thresholds for mAP are set, namely, $\text{mAP}_{0.5}$ and $\text{mAP}_{0.5:0.95}$, representing the specific values of mAP at different IoU thresholds. The number of parameters is used to evaluate the training and deployment costs while the FPS is employed to assess the inference speed of the models.

2) Experimental Environment and Parameter Settings:

The experiments are carried out on a system equipped with Windows 11 (Professional Edition), featuring an Intel Core i9-13900K processor, 64 GB of RAM, and an RTX 4090 GPU with 24 GB of memory. The experimental framework environment consists of PyTorch 1.31.1, Python interpreter version 3.8.18, and CUDA version 11.7.

Our experiments are trained and validated using the same hyperparameters to ensure fairness. The experiments are conducted in a consistent experimental environment, and no pretrained weights are utilized during the training process of any models in the ablation studies or comparative experiments. Improvements are made based on the original models, with each model being iteratively updated 300 times. Each set of experiments is repeated four times and the values reported in the experimental tables represent the averages of these four trials. The hyperparameters for the experiments are set as follows: the total number of training epochs is set at 300, the initial learning rate is set at 0.0001, the batch size, which determines the number of samples input to the model at once, is set at 8, the number of processes is set at 4 and the input image size is set to 640×640 pixels. AdamW optimizer is used to adjust the learning rate during model training. To avoid overfitting, a weight decay strategy is used, with the weight decay value set at 0.0001.

B. Ablation Experiments

1) *Module Ablation Experiments:* To evaluate the contribution of each module to the model, ablation experiments are conducted on the SIMD, DOTA-v1.0, and NWPU VHR-10 datasets, using the baseline model RT-DETR for comparison. The baseline model is denoted as Experiment A. Subsequently, Faster-CGSU, CGA-AIFI, CDFF, and Inner-MPDIOU are added to the baseline model in sequence, resulting in Experiments B, C, D, and E, respectively.

Under identical experimental conditions, the experimental results are presented in Table I. In the ablation experiment B of the SIMD dataset, when the basic block is replaced by the Faster-CGSU module, the parameter count of the model is reduced to 15.4 M while the $\text{mAP}_{0.5}$ and $\text{mAP}_{0.5:0.95}$ improved by 1.2% and 0.6%, respectively, due to its feature extraction capability and elimination of redundancy parameters. In Experiment C, after introducing CGA-AIFI, its CGA mechanism effectively improves the detection accuracy for small, medium, and large objects with parameter count further reduced to 15.1 M and P and R increase by 0.5% and 0.3%, respectively. However, the improvements in $\text{mAP}_{0.5}$ and AP_s were relatively small. This is because CGA-AIFI focuses primarily on high-level semantic information with larger receptive fields that can prevent the loss of global

information, making it less sensitive to small objects, especially with sparse distributed features. In Experiment D, after the introduction of CDFF, AP_s , AP_m , and AP_l increased by 0.8%, 0.8%, and 2.4%, respectively. This indicates that CDFF can effectively enhance the performance and robustness of the model against object scale variations through multiscale feature alignment, fine-grained enhancement, and dynamic fusion strategies. In Experiment E, our Inner-MPDIOU loss function makes all metrics grow simultaneously compared to Experiment D, with AP_s further increasing by 1.7% and FPS increasing by 10.4 compared to Experiment A, effectively improving the model's ability to focus on difficult small objects.

In the DOTA-v1.0 dataset, after introducing the Faster-CGLU module in Experiment B, AP_s , AP_m , and AP_l increased by 0.4%, 0.5%, and 0.7%, respectively, compared to the baseline model, while also accelerating the inference speed of the model with a lower number of parameters. This demonstrates that the CGSU within Faster-CGSU can more effectively capture local features, enhancing the model's ability to process subtle characteristics. In Experiment C, the CGA-AIFI module was introduced to expand the model's receptive field, assisting the model in better balancing local and global features to improve its perception of objects at different scales. AP_s increased by 0.7%, while AP_m and AP_l increased by 0.7% and 0.5%, respectively, compared to the baseline model. In Experiment D, after the introduction of CDFF, the AP_s increased by 1.6% compared to the baseline, which is attributed to DSSFF within CDFF, employing the channel-parallel dynamic upsampling strategy to adaptively generate sampling points based on image content, thus mitigating the loss of small object feature information. AP_m increases by 0.9%, indicating that the TFE focuses more on medium-scale characteristics, preserving the integrity of the feature structures. Meanwhile, the 3D convolution in DSSFF effectively constructs the semantic continuity between features at different scales, enhancing the perception of the model to structural information. AP_l increased by 1.0%, indicating that the CDFF module can dynamically fuse deep semantic information with shallow details, improving the general representation and localization accuracy of large objects. In Experiment E, after introducing the Inner-MPDIOU loss function, the detection accuracy for small, medium, and large objects is further improved, with AP_s increasing by 2.2% compared to the baseline.

In the NWPU VHR-10 dataset, when comparing Experiment E with Experiment A, P, $\text{mAP}_{0.5}$, and $\text{mAP}_{0.5:0.95}$ improved by 4.2%, 3.1%, and 2.9%, respectively, with other metrics also reaching optimal levels. This shows that the HPS-DETR can achieve a higher detection accuracy and verifies the excellent generalization performance.

The ablation experiments collectively demonstrate that the four enhancements introduced to RT-DETR have resulted in improvements in both mAP and mAP_s , along with an average of 10 frames increase in real-time performance compared to the baseline model, while simultaneously reducing the parameters of the model.

TABLE I

QUANTITATIVE EVALUATION OF THE PERFORMANCE OF EACH MODULE OF HPS-DETR ON SIMD, DOTA-v1.0, AND NWPU VHR-10 DATASETS

Name	Method	Params/M	SIMD [23]							
			P/%	R/%	mAP _{0.5} /%	mAP _{0.5:0.95} /%	AP _s /%	AP _m /%	AP _l /%	FPS
A	Baseline	19.9	77.9	77.2	77.4	62.0	25.4	47.9	65.4	70.3
B	A+Faster-CGSU	15.4	78.1	78.2	78.6	62.6	25.1	48.6	68.1	78.7
C	B+CGA-AIFI	15.1	78.4	77.5	78.1	62.1	25.7	49.1	67.2	79.2
D	C+CDFF	15.5	81.2	77.9	79.5	63.2	26.2	48.7	68.2	77.2
E	D+Inner-MPDIoU	15.5	82.8	78.4	79.8	63.5	27.9	49.2	68.5	80.7
Name	Method	Params/M	DOTA-v1.0 [24]							
			P/%	R/%	mAP _{0.5} /%	mAP _{0.5:0.95} /%	AP _s /%	AP _m /%	AP _l /%	FPS
A	Baseline	19.9	85.7	80.9	85.1	60.7	37.0	64.1	67.6	68.8
B	A+Faster-CGSU	15.4	86.2	81.5	85.7	60.9	37.4	64.6	68.3	65.2
C	B+CGA-AIFI	15.1	85.9	79.9	85.4	61.2	38.1	64.2	68	74.8
D	C+CDFF	15.5	88.3	81.8	85.3	61.8	38.6	65.0	68.6	79.4
E	D+Inner-MPDIoU	15.5	89.4	82.3	86.8	62.5	39.2	65.4	69.2	80.2
Name	Method	Params/M	NWPU VHR-10 [25]							
			P/%	R/%	mAP _{0.5} /%	mAP _{0.5:0.95} /%	AP _s /%	AP _m /%	AP _l /%	FPS
A	Baseline	19.9	88.5	85.2	88.4	57.5	26.5	52.5	65.2	74.3
B	A+Faster-CGSU	15.4	87.6	83.4	87.7	57.4	27.4	54.3	63.4	70.9
C	B+CGA-AIFI	15.1	89.9	87.6	89.4	58.3	28.6	55.3	63.6	75.7
D	C+CDFF	15.5	91.6	86.9	90.6	58.8	29.2	58.9	65.8	82.4
E	D+Inner-MPDIoU	15.5	92.7	87.8	91.5	60.4	29.8	59.6	66.9	82.8

TABLE II

PRECISION EVALUATION OF HPS-DETR MODULES FOR DIFFERENT DETECTION OBJECTS ON SIMD DATASET

Faster-CGSU	CGA-AIFI	CDFF	Inner-MPDIoU	AP/%												mAP /%
				C	T	V	LV	B	PA	TA	FA	ST	PT	H	S	
×	×	×	×	90.2	77.8	78.4	83.3	89.4	71.9	94.8	98.4	50.3	40.2	64.8	95.7	77.9
✓	×	×	×	90.2	77.3	77.6	83.2	92.8	76.7	93.4	99.5	56.4	44.5	65.9	96.4	79.5
✓	✓	×	×	89.4	75.5	78.5	83.4	91.1	75.8	93.5	98.7	53.6	41.2	84.0	96.3	80.1
✓	✓	×	✓	89.8	76.5	78.9	83.6	92.3	76.1	94.4	99.3	52.8	44.8	83.6	96.4	80.6
✓	×	✓	×	90.4	76.6	78.1	83.5	91.3	76.8	94.8	98.5	51.3	44.9	80.5	95.8	80.2
✓	×	✓	✓	90.3	77.5	78.7	83.3	92.6	77.1	94.7	99.0	51.8	45.3	83.0	96.1	80.7
×	✓	✓	×	90.3	76.0	78.6	83.5	90.1	75.9	94.6	99.5	50.6	45.1	75.2	96.6	79.6
×	✓	✓	✓	90.2	78.1	79.1	83.4	93.0	76.4	94.3	99.5	53.1	46.4	77.3	96.5	80.6
✓	✓	✓	×	90.6	78.6	80.4	83.7	92.7	76.2	94.7	99.6	55.7	48.8	78.3	96.3	81.3
✓	✓	✓	✓	90.5	78.8	79.2	83.5	92.6	76.5	94.9	99.5	58.1	47.6	83.9	96.2	81.8

To assess the influence exerted by each module within HPS-DETR on diverse categories of objects, we perform experimental validation on a selection of 12 relatively smaller object classes from the SIMD dataset, encompassing C, T, V, LV, B, PA, TA, FA, ST, PT, H, and S. The results are presented in Table II, where “✓” indicates that the module is activated and “×” denotes its deactivation. The detection accuracy for each category of small objects has been improved, with particularly high adaptability for categories such as T, TA, FA, and S. The mAP for all small object categories has increased by 3.9% compared to the baseline. However, the accuracy improvement for some objects is relatively minor, mainly due to their fuzzy boundaries or the presence of fine-grained local features. The model exhibits lower detection accuracy for PT and ST, with accuracies of 47.6% and 58.1%, respectively. In addition to the influencing factors mentioned

above, this is also attributed to their susceptibility to occlusion. However, compared to the baseline, the accuracies of PT and ST have increased by 7.4% and 7.8%, respectively. Therefore, our HPS-DETR significantly enhances the detection capability of small objects occluded.

2) *Loss Function Ablation Experiments:* To evaluate the performance of Inner-MPDIoU, we conduct 16 sets of experiments on the SIMD dataset and the NWPU VHR-10 dataset, respectively, comparing the effect of different IoUs on our model performance. The experimental results are presented in Table III. When HPS-DETR utilized the loss function of the baseline, GIoU, it achieved detection accuracies of 79.5% and 90.6% in terms of mAP_{0.5}, respectively. Although these results demonstrate high accuracy, they are still inferior to those obtained with Inner-MPDIoU. Then, a comparison is implemented across five sets of experiments with varying

TABLE III
PERFORMANCE EVALUATION TABLE OF DIFFERENT IoU METRICS ON SIMD AND NWPU VHR-10 DATASETS

Loss Function		SIMD [23]				NWPU VHR-10 [25]			
		P/%	R/%	mAP _{0.5} /%	mAP _{0.5:0.95} /%	P/%	R/%	mAP _{0.5} /%	mAP _{0.5:0.95} /%
GIoU [57]		81.2	77.9	79.5	63.2	91.6	86.9	90.6	58.8
CIoU [58]		81.4	78.2	79.2	62.9	91.8	87.6	91.2	58.9
DIOU [59]		79.3	77.8	78.7	62.4	88.5	88.2	91.0	59.2
EIoU [60]		80.4	76.7	78.0	62.2	90.6	86.6	89.2	57.8
SIoU [61]		82.1	76.6	79.1	62.9	91.8	84.7	89.9	58.6
MPDIoU [56]		81.6	78.7	79.8	63.1	88.8	88.0	90.8	58.8
Inner-IoU [55]	ratio=0.7	78.7	77.3	77.0	61.8	90.0	88.4	89.4	58.6
	ratio=0.75	79.3	78.6	78.4	62.1	90.6	85.5	89.6	59.6
	ratio=0.85	79.8	76.9	78.2	62.2	90.7	85.9	90.5	59.8
	ratio=0.95	81.0	76.1	77.6	62.1	91.2	88.4	91.2	60.1
	ratio=1.0	78.2	76.0	77.9	61.9	89.5	86.1	89.6	57.3
Inner-MPDIoU	ratio=0.7	80.6	78.9	79.4	62.9	87.8	88.1	89.3	58.3
	ratio=0.75	81.8	77.6	79.0	63.2	88.3	87.6	90.1	58.7
	ratio=0.85	80.3	78.4	79.0	62.8	89.9	88.5	91.0	58.9
	ratio=0.95	82.8	78.4	79.8	63.5	92.7	87.8	91.5	60.4
	ratio=1.0	79.0	77.9	78.2	62.4	90.1	87.3	90.2	58.3

TABLE IV
PERFORMANCE EVALUATION OF THE SCALING COEFFICIENT IN THE DYNAMIC SAMPLING FACTOR OF CP-DYSAMPLE

Sampling Factor	P/%	R/%	AP _s /%	AP _m /%	AP _l /%
0.1 δ	89.7	85.4	23.9	54.9	63.9
0.3 δ	90.8	86.3	28.4	56.1	65.7
0.5 δ	92.7	87.8	29.8	59.6	66.9
0.7 δ	91.5	87.2	24.7	57.9	65.8
0.9 δ	90.4	86.0	23.2	56.2	64.6

scaling factor (ratio) for both Inner-IoU and Inner-MPDIoU. Inner-IoU is observed to achieve the highest precision in both datasets when the ratio is set at 0.95. For Inner-MPDIoU with a ratio of 0.95, all metrics reach their optimum. This validates the superiority of Inner-MPDIoU, demonstrating its enhanced ability to focus on dense small objects.

3) *Ablation Experiments on the Scaling Coefficient in the Dynamic Sampling Factor of CP-Dysample*: To evaluate the impact of the constant scaling factor applied before the dynamic sampling factor in the CP-DySample module on the quality of dynamic upsampling, we conducted five groups of ablation experiments on the NWPU VHR-10 dataset using step sizes of 0.2. As shown in Table IV, when the sampling factor is 0.5 δ , all metrics get their optimal values. In addition, when the factor is small, the model cannot flexibly adjust the sampling positions based on the image features, resulting in the loss of small object features. In contrast, the sampling points shift excessively, compromising local feature consistency.

C. Comparison Experiments

1) *Comparison of Different Network Models*: To further verify the superiority of HPS-DETR in remote sensing image object detection tasks, we compare it with classical models in the SIMD, DOTA-v1.0, and NWPU VHR-10 datasets, considering both the parameters and precision

metrics. The algorithms selected for comparison include Faster-RCNN [9], SSD [10], EfficientDet [62], YOLOv3 [63], YOLOv5, YOLOv6 [64], YOLOv8, YOLOv9 [65], YOLOv10 [66], YOLOv11 [67], YOLOv12 [68], Deformable-DETR [15], and DEIM [69].

The experimental results, as delineated in Table V, reveal that Faster-RCNN, SSD, and EfficientDet exhibit comparatively lower precision. YOLOv3 exhibits a marked promotion in detection precision compared to SSD and EfficientDet, but the parameters constrain its practical applicability in real scenarios. The widely used YOLOv5, available in various versions, also displays moderate accuracy on datasets such as SIMD and DOTA-v1.0, which are known for the abundance of small objects. In contrast, YOLOv5l attains a detection accuracy of 91.1% on the NWPU VHR-10 dataset, which has a relatively lower density of small objects. YOLOv8, offered in various size configurations, demonstrates superior performance. YOLOv8m and YOLOv8l possess larger parameters than HPS-DETR and lag behind in detection accuracy. Recent real-time detection models, YOLOv9c and YOLOv10m, achieve detection accuracies of 78.6% and 73.7%, respectively, on the DOTA-v1.0 dataset, still slightly inferior to HPS-DETR, and remain substantial scope for enhancing the detection accuracy of small objects in remote sensing imagery. The latest YOLOv11m and YOLOv12m models show slightly lower detection accuracy compared to HPS-DETR in all three datasets, while having larger parameters.

The incorporation of the multihead attention mechanism in transformer architectures, such as Deformable-DETR, baseline, and RT-DETR (r50), contributes to their higher detection accuracy. However, their significantly higher parameter counts make them challenging to deploy in practice compared to ours. Compared with the latest real-time SOTA model, DEIM, HPS-DETR performs well both in parameters and precision, achieving mAP_{0.5} improvements of 0.7%, 1.5%, and 0.8% in

TABLE V
PERFORMANCE COMPARISON OF DIFFERENT NETWORK MODELS ON SIMD, DOTA-v1.0, AND NWPU VHR-10 DATASETS

Method	Params/M	SIMD [23]		DOTA-v1.0 [24]		NWPU VHR-10 [25]	
		mAP _{0.5} /%	mAP _{0.5:0.95} /%	mAP _{0.5} /%	mAP _{0.5:0.95} /%	mAP _{0.5} /%	mAP _{0.5:0.95} /%
Faster-RCNN [9]	136.9	62.4	42.1	42.3	25.4	78.4	40.3
SSD [10]	26.3	59.8	40.8	59.6	37.8	72.5	38.4
EfficientDet [62]	12.0	56.1	40.2	55.7	34.7	70.3	38.0
YOLOv3 [63]	103.7	60.3	41.8	68.2	45.6	84.4	51.6
YOLOv5m	21.6	69.3	44.3	69.4	46.2	83.4	50.8
YOLOv5l	46.2	74.5	58.6	71.3	49.5	91.1	56.9
YOLOv6l [64]	20.3	72.8	55.7	70.6	47.5	90.2	57.3
YOLOv8s	11.7	76.5	60.4	68.9	45.8	89.6	57.0
YOLOv8m	25.8	77.2	61.3	71.2	49.8	90.9	57.8
YOLOv8l	43.6	77.9	61.8	74.8	53.6	91.3	58.2
YOLOv9m [65]	20.1	76.5	61.2	73.5	52.7	89.8	59.4
YOLOv9c [65]	25.3	78.2	62.3	78.6	56.3	90.3	60.1
YOLOv10m [66]	16.4	76.3	61.2	73.7	53.3	90.0	59.8
YOLOv11m [67]	20.1	78.8	62.4	82.1	60.2	89.7	57.6
YOLOv12m [68]	20.2	77.9	62.0	86.2	62.0	86.6	53.1
Deformable-DETR [15]	40.0	75.2	59.4	78.4	56.5	87.6	56.3
DEIM [69]	19.2	79.1	63.0	85.3	61.2	90.7	59.3
RT-DETR (r50) [22]	42.9	78.3	62.7	85.9	61.4	89.8	58.2
Baseline [22]	19.9	77.4	62.0	85.1	60.7	88.4	57.5
HPS-DETR (ours)	15.5	79.8	63.5	86.8	62.5	91.5	60.4

TABLE VI
PRECISION EVALUATION OF DIFFERENT NETWORK MODELS FOR VARIOUS DETECTION OBJECTS ON DOTA-v1.0 DATASET

Method	AP/%													mAP/%
	PL	SH	ST	BD	TC	BC	HA	BR	LV	SV	HC	RA	SP	
Faster-RCNN [9]	52.8	35.6	42.5	65.3	52.7	45.4	42.6	38.9	39.4	28.6	42.8	32.7	47.7	43.6
SSD [10]	79.1	46.9	53.3	78.6	80.7	52.7	49.4	38.2	53.6	42.2	60.2	47.5	72.6	58.1
EfficientDet [62]	68.5	43.2	50.1	73.5	72.3	68.6	55.4	30.5	55.6	34.8	55.4	38.9	60.8	54.4
YOLOv3 [63]	83.5	78.6	62.2	54.7	78.6	56.7	79.8	29.7	71.2	68.4	79.6	44.9	78.8	66.7
YOLOv5m	88.3	76.5	85.2	59.5	89.4	62.7	82.6	44.2	75.8	69.0	57.1	66.0	57.8	70.3
YOLOv5l	92.0	87.5	70.9	68.4	94.1	64.6	85.1	49.6	85.6	66.9	59.7	53.0	62.7	72.3
YOLOv6l [64]	90.1	87.3	72.7	65.8	83.2	68.7	79.7	46.2	76.2	79.4	57.5	42.6	65.8	70.4
YOLOv8s	90.6	86.5	77.1	56.8	94.5	70.3	85.3	40.8	78.3	74.7	53.6	45.8	63.7	70.6
YOLOv8m	92.3	85.1	78.5	47.3	93.2	69.7	73.4	39.8	83.5	81.6	74.6	43.8	75.3	72.1
YOLOv8l	92.2	86.7	79.6	45.6	94.6	82.3	75.6	35.4	82.7	81.4	75.5	42.6	80.1	73.4
YOLOv9m [65]	95.8	83.2	69.5	55.8	93.5	88.6	80.0	38.4	85.5	82.7	78.6	41.2	86.4	75.3
YOLOv9c [65]	94.8	90.8	74.2	69.2	92.5	69.9	87.7	56.6	90.8	81.5	75.5	58.5	88.8	79.2
YOLOv10m [66]	93.5	87.6	63.5	83.2	88.7	71.6	89.3	45.2	78.7	75.9	74.8	40.8	78.6	74.7
YOLOv11m [67]	97.0	91.0	84.3	82.0	92.5	89.9	89.5	61.3	87.5	84.7	79.2	66.4	90.2	83.5
YOLOv12m [68]	96.8	91.5	85.0	83.4	93.6	90.1	88.5	60.6	89.5	84.6	77.4	68.0	88.9	85.3
Deformable-DETR [15]	96.3	75.9	58.6	68.7	88.4	70.5	65.3	49.5	91.7	88.6	71.3	50.3	73.4	72.9
DEIM [69]	96.6	90.1	81.3	80.8	96.4	84.1	88.2	59.3	92.9	88.7	92.4	69.3	92.7	85.6
RT-DETR (r50) [22]	95.4	89.3	80.6	75.6	98.3	78.9	86.7	54.9	92.6	84.5	84.3	56.4	93.1	82.4
Baseline [22]	96.8	88.8	81.5	81.7	96.3	91.8	88.1	69.0	90.6	87.8	89.3	65.2	91.5	86.0
HPS-DETR (ours)	97.3	88.9	83.2	82.3	95.8	91.1	88.7	72.2	91.6	89.7	92.9	72.2	92.1	87.5

the SIMD, DOTA-v1.0, and NWPU VHR-10 datasets, respectively. HPS-DETR achieves a substantial leap in detection accuracy, attaining the highest level. With a mere parameter count of 15.5 M, it achieves a mAP_{0.5} accuracy of 79.8%, 86.8%, and 91.5% in these datasets, respectively. Furthermore, the complete precision, mAP_{0.5:0.95}, reaches 63.5%, 62.5%, and 60.4%, respectively. In comparison to other mainstream algorithms, HPS-DETR attains superior performance, especially in precision with its lightweight design.

Furthermore, given the extensive diversity of object categories, feature information and scale within the DOTA-v1.0, we select 13 specific categories of smaller objects which we focus on for analysis: PL, SH, ST, BD, TC, BC, HA, BR, LV, SV, HC, RA, and SP. The detection results are presented in Table VI. Compared to the main algorithms, HPS-DETR shows superior precision performance, achieving an mAP of 87.5% on the DOTA-v1.0 dataset. In particular, it exhibits exceptional proficiency in detecting five particularly

TABLE VII

PERFORMANCE COMPARISON OF DIFFERENT LIGHTWEIGHT BACKBONES ON SIMD DATASET

Backbone	Param/M	P/%	R/%	mAP _{0.5} /%	FPS
ResNet18 [70]	20.0	80.1	78.1	78.8	70.1
EfficientViT [53]	10.9	76.7	74.8	75.3	93.5
MobileNetV4 [71]	11.5	78.3	75.3	76.1	90.2
LSKNet [32]	12.7	78.0	76.3	78.2	88.3
FasterNet [51]	16.9	79.2	77.8	77.9	78.4
Faster-CGSU (ours)	15.5	82.8	78.4	79.8	80.7

TABLE VIII

PERFORMANCE COMPARISON OF DIFFERENT ATTENTION MECHANISMS ON DOTA-v1.0 DATASET

Attention	Param/M	P/%	R/%	mAP _{0.5} /%	FPS
MHSA [13]	15.8	87.6	80.0	84.8	74.5
DA [72]	15.9	87.7	80.1	85.5	67.6
M2SA [73]	15.9	86	81.4	85.3	73.8
EAA [74]	16.0	87.8	80.4	85.5	65.4
HiLo [75]	15.7	87.5	81.1	86.0	76.6
CGA-AIFI (ours)	15.5	89.4	82.3	86.8	80.2

challenging categories: PL, BR, SV, HC, and RA. The detection accuracy for three categories of small objects surpasses that of YOLO-based models by a significant margin: PL, LV, and SV. Consequently, it is evident that HPS-DETR achieves remarkably notable success in small object detection.

Compared to the SOTA model, DEIM, HPS-DETR outperforms in more than half of the object categories, demonstrating its remarkable performance in the detection of small objects. For large objects such as BD and BC, HPS-DETR also exhibits excellent precision performance, demonstrating that it maintains sensitivity to large object detection while improving the detection accuracy of small objects.

2) *Comparison of Different Lightweight Backbones:* To validate the effectiveness of Faster-CGSU, the lightweight feature extraction module in the backbone of HPS-DETR, we select five current main lightweight feature extraction networks: ResNet18 [70], EfficientViT [53], MobileNetV4 [71], LSKNet [32], and FasterNet [51], replace the HPS-DETR backbone, and perform comparative experiments on the SIMD dataset. The experimental results are presented in Table VII. It can be seen that although EfficientViT, MobileNetV4, and LSKNet have fewer parameters, they perform poorly in terms of P, R, and mAP_{0.5}, indicating their limited ability to extract small objects or detailed features in complex remote sensing images. ResNet18 and FasterNet, with larger parameter counts, exhibit significantly lower inference speeds (FPS). In contrast, Faster-CGSU achieves the best in P, R, and mAP_{0.5}, with a moderate number of parameters and inference speed. Given the practical application requirements of detecting as many remote sensing objects as possible with high speed and minimal omissions, Faster-CGSU is chosen as the optimal option, achieving a good balance between accuracy and efficiency in the feature extraction stage.

3) *Comparison of Different Attention Mechanisms:* To further validate the superiority of cascaded grouped attention used in CGA-AIFI, we conduct comparative experiments on the

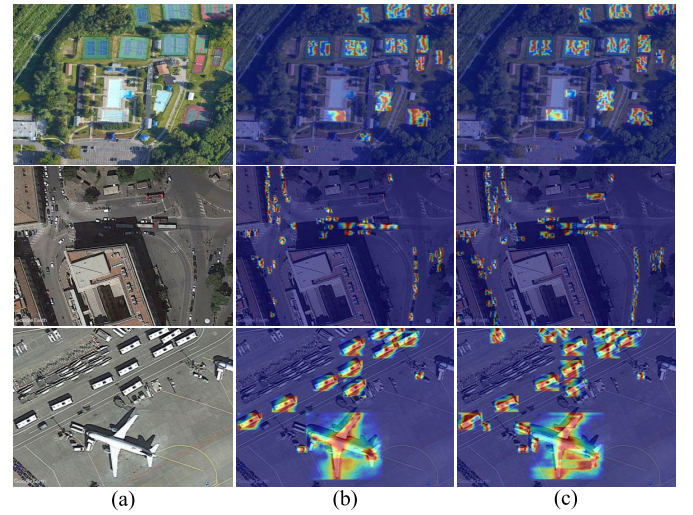


Fig. 8. Comparison of heatmaps between the baseline model and ours. (a) Original image. (b) Heatmap of baseline. (c) Heatmap of HPS-DETR. HPS-DETR demonstrates significant advantages, being able to detect occluded and small objects that baseline fail to identify.

DOTA-v1.0 dataset, which contains complex backgrounds and a large number of densely distributed small objects. The compared attention mechanisms include MHSA [13], deformable attention (DA) [72], multiscale MHSA (M2SA) [73], effective additive attention (EAA) [74], and HiLo attention (HiLo) [75]. The experimental results are shown in Table VIII. It can be seen that with the introduction of the CGA-AIFI module, HPS-DETR achieved the best performance across all evaluation metrics while maintaining the lowest parameter count among the five attention mechanisms. Furthermore, CGA-AIFI achieved an FPS of 80.2, significantly higher than EAA and DA, indicating its lower computational cost while maintaining high accuracy. The cascaded and group strategies of CGA-AIFI enable a progressive fusion of feature information from local to global contexts, significantly improving the ability of the model to perceive small objects in complex backgrounds.

D. Visualization Experiments

1) *Model Heatmaps:* To intuitively demonstrate the detection performance of HPS-DETR, we employ the gradient-weighted class activation mapping (Grad-CAM) visualization technique, which is used to illustrate the contributions of feature maps within deep neural networks to prediction performance. Pixels with higher gradients in the feature maps are represented by darker red in the heatmaps, indicating greater attention to those areas, and pixels with lower gradients are depicted in darker blue, signifying reduced sensitivity to those regions. The experimental results are presented in Fig. 8 and it can be concluded that, in the context of remote sensing object detection, the baseline RT-DETR exhibits lower attention to small objects and reduced sensitivity to objects in complex environments. In contrast, HPS-DETR demonstrates significant advantages by successfully detecting small and obscured objects that the baseline fails to identify. It accurately detects objects of varying scales and different categories,

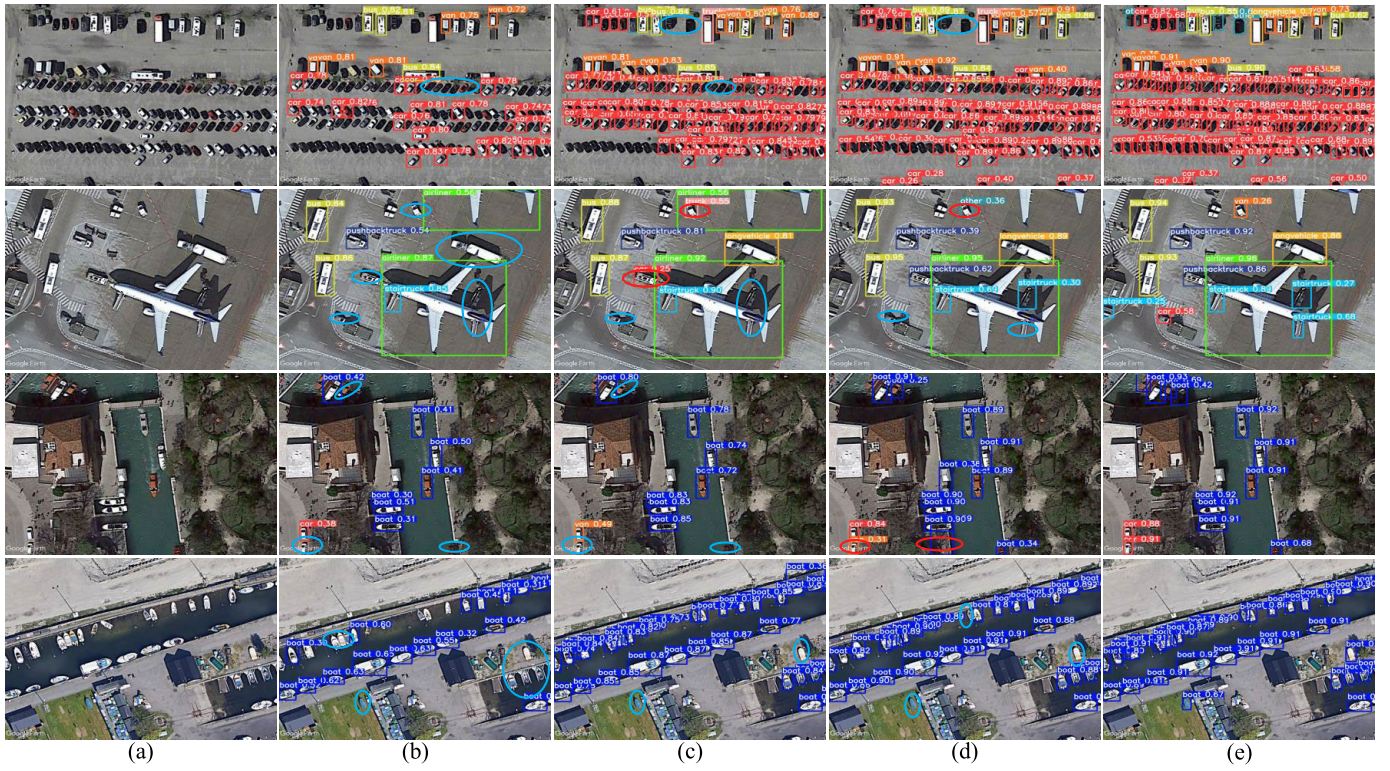


Fig. 9. Comparison of detection performance of different models on different objects in SIMD. (a) Original image. (b) EfficientDet detection results. (c) YOLOv8m detection results. (d) Baseline detection results. (e) HPS-DETR detection results. The blue ellipses represent missed objects, and the red ellipses represent false positives.

further validating the effectiveness of the proposed improvements in enhancing the precision performance to smaller object detection, providing more precise information extraction capabilities for relevant applications.

2) *Detection Results*: To gain a more vivid appreciation of the detection performance of HPS-DETR in various interference environments, different scales, and numerous small objects in remote sensing images, we conduct a qualitative comparative experiment involving several benchmark models. Specifically, we select the classical EfficientDet, the currently widespread YOLOv8m, and the baseline RT-DETR for comparison.

To validate the detection performance, we randomly select four types of SIMD images that featured dense small objects, multiscale objects, and complex backgrounds. The detection results are presented in Fig. 9. It can be seen that EfficientDet detected the least number of objects and exhibited lower classification confidence scores (column b). YOLOv8m demonstrated a significant improvement in the detection of objects in different scenarios. However, in the top and bottom images of column c, which have small and dense objects, a considerable number of objects are still missed. In scenes with multiscale objects (the 2nd row in column c), YOLOv8m misidentified aircraft trailers as cars and misclassified vans as trucks, indicating its limitation in detecting objects of varying categories and sizes. The baseline, RT-DETR, exhibits relatively balanced performance in various scenarios, particularly in scenes with densely arranged and multiscale objects (the 1st and 2nd rows of column d), where it could detect most types of vehicles with higher classification confidence scores.

However, it still suffers from missed and false detections when faced with complex backgrounds and numerous small objects (the 3rd and 4th rows of column d). However, HPS-DETR accurately identifies objects in almost all scenarios (column e), successfully detecting dense and small objects, and demonstrating its adaptation to remote sensing scenes. This indicates that HPS-DETR possesses strong feature extraction and precise localization capabilities with robust scene adaptability.

To evaluate the detection performance of the models, we randomly select two types of classic images featuring dense and multiscale objects from NWPU VHR-10, as well as two types of classic images with complex backgrounds and numerous small objects from DOTA-v1.0. The detection results are illustrated in Fig. 10. It is evident that among the four types of remote sensing image object detection tasks, EfficientDet and YOLOv8m exhibit poorer detection performance, with a significant number of missed detections. This can be primarily attributed to inadequate feature extraction and fusion. The baseline, RT-DETR, demonstrates moderate detection performance. In scenes with dense and numerous small objects, it still has some missed detections of small objects. In contrast to those, HPS-DETR achieves the best detection performance. It accurately identifies objects either on water or on land, with higher confidence scores. This is attributed to the combination of efficient feature extraction of the proposed Faster-CGSU module, fine-grained queries with CGA-AIFI, deep feature fusion through CDFF, and effective focusing on difficult objects by Inner-MPDIoU loss function.

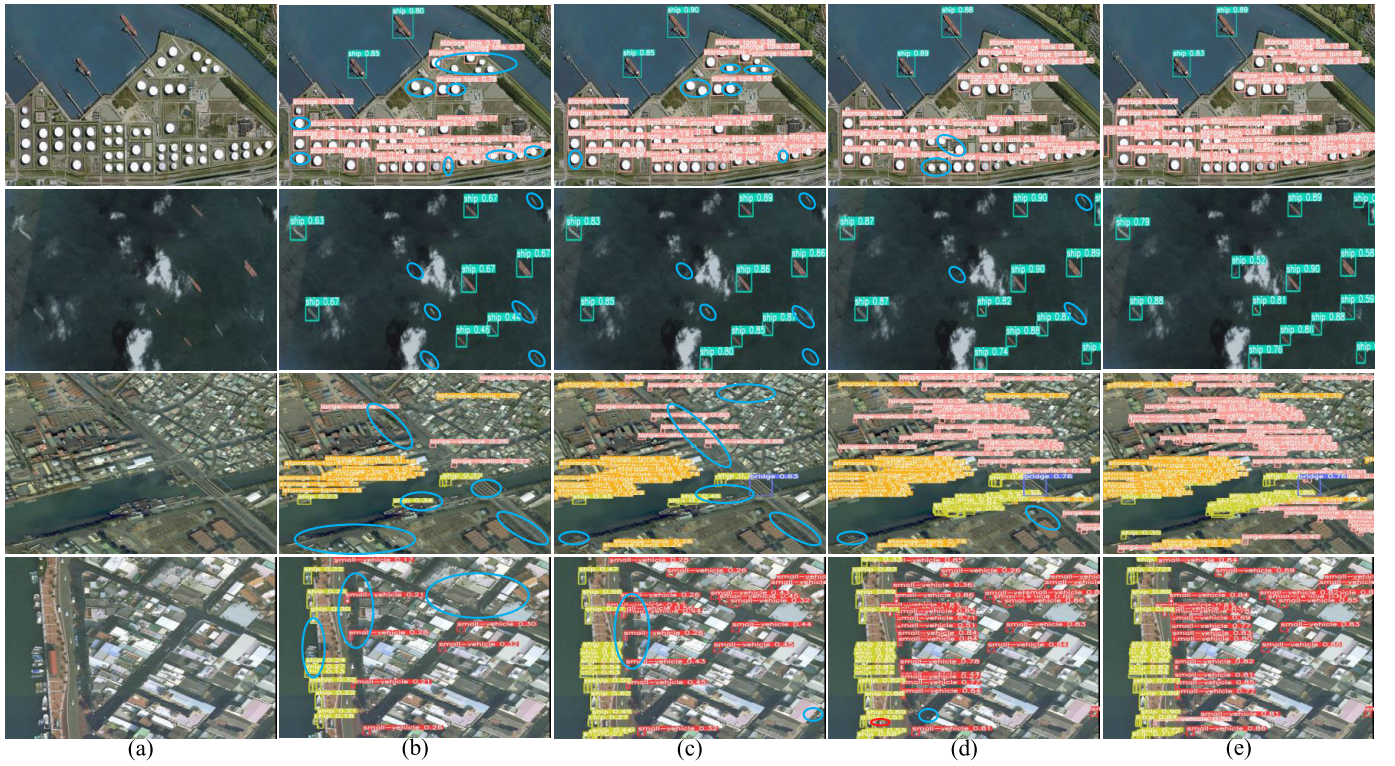


Fig. 10. Comparison of detection performance of different models on different objects in NWPU VHR-10 and DOTA-v1.0. (a) Original image. (b) EfficientDet detection results. (c) YOLOv8m detection results. (d) Baseline detection results. (e) HPS-DETR detection results. The blue ellipses represent missed objects, and the red ellipses represent false positives.

TABLE IX

GENERALIZATION EVALUATION OF THE BASELINE AND HPS-DETR ON WEATHERDRONE

Method	Param/M	P/%	R/%	mAP _{0.5} /%	FPS
Baseline	19.9	56.9	41.6	42.3	72.3
HPS-DETR	15.5	58.8	42.9	44.2	81.4

TABLE X

GENERALIZATION EVALUATION OF BASELINE AND HPS-DETR ON VEDAI

Method	Param/M	P/%	R/%	mAP _{0.5} /%	FPS
Baseline	19.9	64.3	57.4	55.6	63.4
HPS-DETR	15.5	67.4	60.3	59.4	73.1

E. Discussion on Model Generalization

To further validate the generalizability of HPS-DETR, experiments are carried out on WeatherDrone [26], VEDAI [27], and RTTS [28] small object datasets, using both the baseline and HPS-DETR. The WeatherDrone dataset incorporates complex weather noises such as rain, snow, and fog, and spans various lighting scenarios, including daytime and nighttime. VEDAI can be utilized to test generalizability in the detection of tiny objects. RTTS is a foggy weather dataset designed for the field of traffic object detection. It features a high proportion of small objects with a 3-D distribution pattern, along with challenges such as dense occlusions and significant scale variations, and can be utilized to validate the generalization ability of algorithms across different domains. All experiments are conducted under the same conditions as those described in previous experiments.

The generalization experiment results on the WeatherDrone dataset are presented in Table IX. Compared to baseline, HPS-DETR reduces the parameters by 4.4 M, while improving P and R by 1.9% and 1.3%, respectively. This indicates a decrease in the missed detection rate by 1.3%, with most of

the missed objects smaller than 10×10 pixels and densely overlapped. Furthermore, HPS-DETR achieved an FPS of 81.4, demonstrating enhanced real-time performance in large-scale scene object detection tasks. The visualization results of the generalization are illustrated in Fig. 11. Compared to baseline, HPS-DETR is capable of detecting small objects densely packed in occluded regions, such as pedestrians, bicycles, and motorcycles. Meanwhile, it demonstrates higher overall confidence in the results of object classification, showcasing stronger robustness and adaptability under varying lighting conditions and complex weather scenarios. However, in snowy conditions, HPS-DETR still encounters certain challenges when detecting small objects with sizes smaller than 10×10 pixels.

In the generalization experiments conducted on VEDAI, Table X shows that HPS-DETR continues to achieve outstanding results. P and R increase by 3.1% and 2.9%, respectively, while mAP_{0.5} improves from 55.6% to 59.4% and comprehensive detection accuracy mAP_{0.5:0.95} increases to 34.2%, showing a significant improvement over the baseline. The image detection results of the two models on VEDAI are

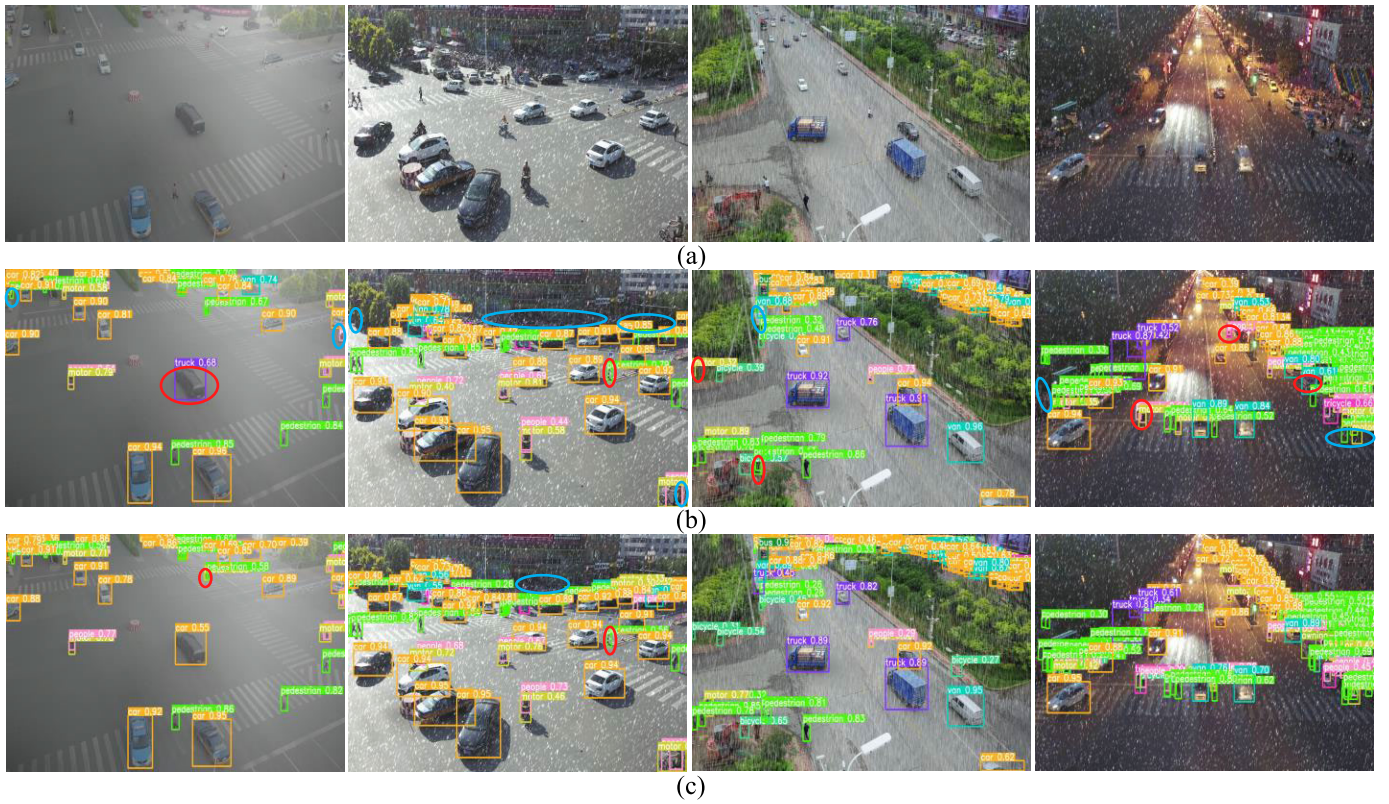


Fig. 11. Detection results of RT-DETR and HPS-DETR on the WeatherDrone dataset under complex weather conditions, and from left to right, it is foggy, snowy, rainy, and snowy nighttime scenes. (a) Original images. (b) Baseline detection results. (c) HPS-DETR detection results. The blue ellipses represent missed objects, and the red ellipses represent false positives.

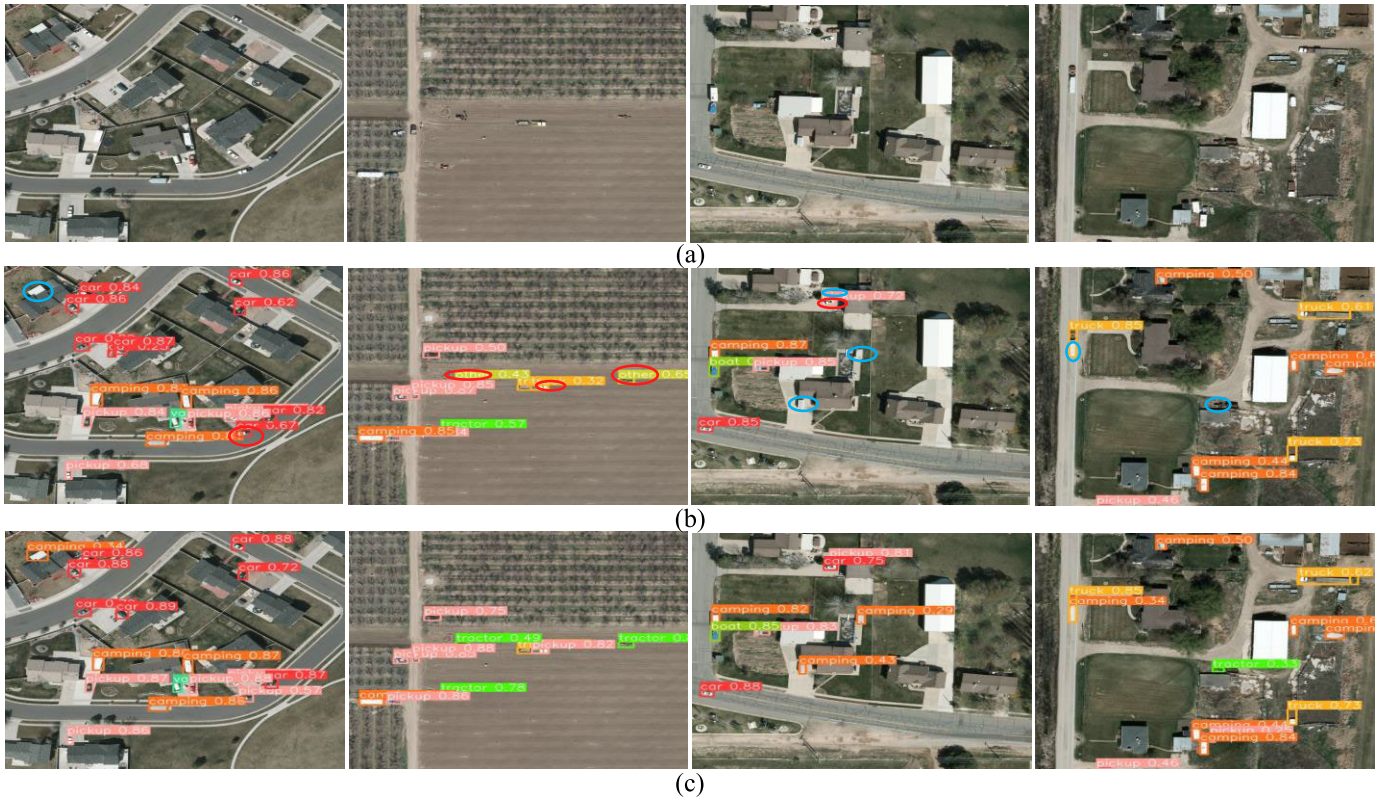


Fig. 12. Detection results of RT-DETR and HPS-DETR on VEDAI. (a) Original image. (b) Baseline detection results. (c) HPS-DETR (Ours) detection results. The blue ellipses represent missed objects, and the red ellipses represent false positives.

shown in Fig. 12. In detecting small objects on various terrains, such as roads, houses, and farmland, HPS-DETR compensates for the shortcomings of the baseline. In particular, in the 2nd row images of farmland, where two tractors, a pickup truck,

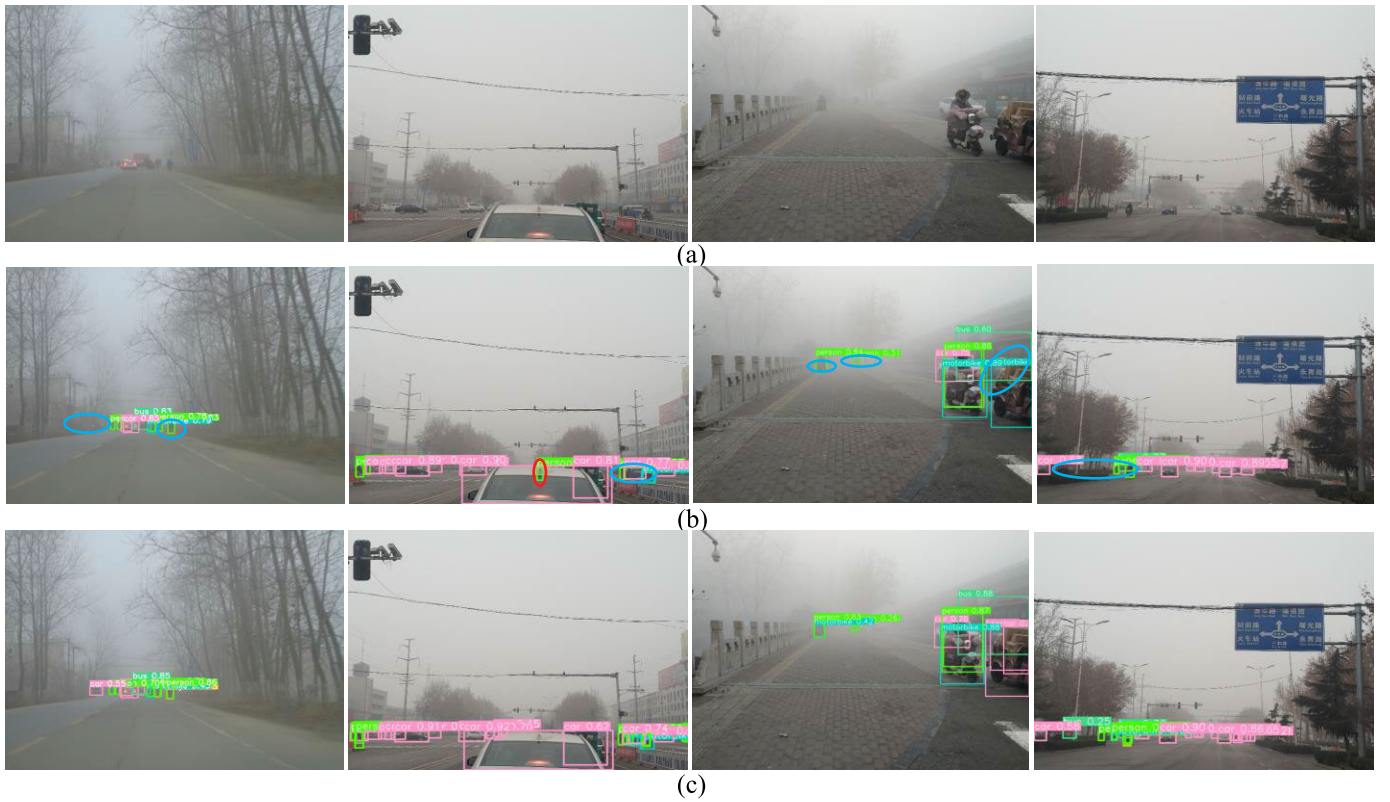


Fig. 13. Detection results of RT-DETR and HPS-DETR on the RTTS. (a) Original images. (b) Baseline detection results. (c) HPS-DETR detection results. The blue ellipses represent missed objects, and the red ellipses represent false positives.

TABLE XI
GENERALIZATION EVALUATION OF THE BASELINE AND HPS-DETR
ON RTTS

Method	Param/M	P/%	R/%	mAP _{0.5} /%	FPS
Baseline	19.9	69.4	54.2	61.4	64.6
HPS-DETR	15.5	72.9	57.3	63.5	75.3

and three tiny objects are misdetected by baseline, HPS-DETR successfully identifies all of them, demonstrating its sensitivity to dense and small objects.

The results of the generalization experiment on the RTTS dataset are shown in Table XI. HPS-DETR demonstrates significant improvements in all detection metrics compared to baseline, particularly with an increase of 2.1% in mAP_{0.5}. Fig. 13 illustrates the detection performance of the two models on traffic objects in actual foggy environments. From the images in the 1st and 4th columns, it can be seen that HPS-DETR is capable of detecting more extremely small objects, such as pedestrians and vehicles. In the images of the 2nd and 3rd columns, HPS-DETR detects a greater number of densely occluded objects. This indicates that HPS-DETR exhibits a high degree of sensitivity toward small objects in the field of traffic object detection and possesses strong generalization capabilities.

In summary, generalization experiments show that regardless of day or night and in various weather conditions such as rain, snow, or fog, HPS-DETR demonstrates exceptional performance, capable of detecting small objects with sizes larger

than 10×10 pixels. However, HPS-DETR still exhibits more missed detections for extremely small and distant objects, especially objects smaller than 10×10 pixels, indicating that it has not yet achieved complete detection ability and still faces certain limitations when dealing with objects with weak feature representations.

V. CONCLUSION

The proposed HPS-DETR integrates advanced modules and loss functions to address the unique challenges of remote sensing object detection, particularly in the context of dense small object distributions with complex backgrounds. Through innovative designs in feature extraction, multiscale feature fusion, and loss function design, HPS-DETR achieves superior detection performance and robustness in remote sensing applications. Extensive experimental results demonstrate that HPS-DETR improves mAP_s by 2.5%, 2.2%, and 3.3% in the SIMD, DOTA-v1.0, and NWPU VHR-10 datasets, respectively. It also increases the average precision across different IoU thresholds (mAP_{0.5:0.95}) by 1.5%, 1.8%, and 2.9%, respectively, while reducing the parameters by 22.11% compared to the baseline. In the WeatherDrone, VEDAI, and RTTS datasets, it achieves mAP_{0.5} of 44.2%, 59.4%, and 63.5%, surpassing the baseline of 1.9%, 3.8%, and 2.1%, respectively. These results show that HPS-DETR outperforms other classical models and presents a more accurate, lightweight, and generalizable solution for small object detection in remote sensing imagery. Experimental results demonstrate

that HPS-DETR achieves a strong detection performance in multiple remote sensing datasets.

However, despite its clear advantages in overall detection accuracy, several challenges remain. There are still missed detections for extremely tiny and relatively distant objects in the image, mainly due to the limited number, weak texture, and irregular shapes of such small objects in the dataset. Future work will explore more discriminative feature enhancement strategies and category-guided feature fusion approaches to address these challenges. Furthermore, although it has been reduced, the parameter size remains high. Future efforts will focus on designing a more lightweight model architecture, with a further increase in detection speed while maintaining competitive accuracy, thus improving applicability in more complex detection scenarios.

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